Fairness-aware PageRank via Edge Reweighting

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Abstract

Link-analysis algorithms, such as PageRank, are instrumental in understanding the structural dynamics of networks by evaluating the importance of individual vertices based on their connectivity. Recently, with the rising importance of responsible AI, the question of fairness in link-analysis algorithms has gained traction.

In this paper, we present a new approach for incorporating group fairness into the PageRank algorithm by reweighting the transition probabilities in the underlying transition matrix. We formulate the problem of achieving fair PageRank by seeking to minimize the fairness loss, which is the difference between the original group-wise PageRank distribution and a target PageRank distribution. We further define a group-adapted fairness notion, which accounts for group homophily by considering random walks with group-biased restart for each group. Since the fairness loss is non-convex, we propose an efficient projected gradient-descent method for computing locally-optimal edge weights. Unlike earlier approaches, we do not recommend adding new edges to the network, nor do we adjust the restart vector. Instead, we keep the topology of the underlying network unchanged and only modify the relative importance of existing edges. We empirically compare our approach with state-of-the-art baselines and demonstrate the efficacy of our method, where very small changes in the transition matrix lead to significant improvement in the fairness of the PageRank algorithm.

CCS Concepts

• **Information systems** → **Social networks**; **Page and site ranking**; **Content ranking**.

Keywords

Link analysis, PageRank, Graph analysis, Fairness

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1 Introduction

Machine-learning algorithms and data-driven decision making are playing increasingly crucial roles in our lives, impacting job markets, healthcare, government, industry, and education. As our dependence on automated decisions grows, however, so does the concern for fairness in algorithms. Ensuring that machine-learning algorithms are reliable and satisfy strong notions of fairness has become an issue of societal concern and, justifiably, has received significant attention in the literature [12, 19]. To date, algorithmic fairness has been studied for primarily supervised [15, 16, 27, 42] but also unsupervised learning [4, 7, 8]. In the context of network and link-analysis algorithms, a few different approaches have been proposed [20, 29, 36, 37, 43], however, the topic has received significantly less attention than the study of fairness in other settings.

Link-analysis algorithms, such as the PageRank family of methods, are fundamental tools for analyzing large-scale network data [3, 14], and have been used in a variety of applications, including identifying authoritative vertices [39], detecting communities [34], uncovering patterns of influence [23], and mitigating network vulnerabilities [18].

Fairness-aware PageRank. The problem of fairness-aware PageRank [37] can be broadly formulated as follows. We are interested in computing the PageRank scores on a graph whose vertices are partitioned into K groups, where the partition is defined by, say, a protected attribute. For simplicity, let us assume $K = 2$ and refer to the two groups of vertices as *red* and *blue*. The fairness of the PageRank distribution is then evaluated based on the total PageRank scores assigned to *red* and *blue* vertices. We would like to avoid situations where the PageRank mass is distributed unevenly (/unfairly) between the two groups. For example, given a target score tuple $\boldsymbol{\phi} = (\phi_{\text{red}}, \phi_{\text{blue}}) = (\phi_{\text{red}}, 1 - \phi_{\text{red}})$, we say that PageRank is $\boldsymbol{\phi}$-fair if the total PageRank score of the vertices in the *red* group is equal to ϕ_{red}; and consequently, the total PageRank score of the vertices in the *blue* group is equal to $1 - \phi_{\text{red}} = \phi_{\text{blue}}$;

Intervention strategies. To improve fairness for PageRank problems, the task is to design an intervention strategy so that the PageRank scores of the graph vertices become $\boldsymbol{\phi}$-fair, or "close enough" to $\boldsymbol{\phi}$-fair. Considering the PageRank algorithm, there are several possible ways to design intervention strategies, including: (*i*) modifying the PageRank restart probability; (*ii*) modifying the restart vector; (*iii*) adding edges to the graph; and (*iv*) modifying the edge weights of the graph in a controlled manner to ensure that the modified graph remains faithful to the original.

In this paper, we focus on intervention strategies of type (*iv*), namely, achieving fair PageRank by reweighting the graph edges. Our motivation is that such interventions are *intrinsic to the graph structure* and can be implemented with actionable operations. For

example, edge reweighting can be achieved by prioritizing/demoting content from one's connections in a social network. Strategies of type (*iii*) involving edge additions or deletions are also intrinsic to the graph structure but tend to be less practical. These strategies rely on link recommendations and have to account for the likelihood of accepting the recommended links, which introduces uncertainty and makes them challenging to implement.

Strategies of type (*i*) and (*ii*), i.e., modifying the PageRank restart probability or restart vector are *extrinsic to the graph structure* and rely on tuning hyper-parameters of the PageRank algorithm. In many cases, it is not meaningful, nor actionable, to modify those hyper-parameters. For example, in the personalized PageRank algorithm, the restart vector has a clear interpretation, which is to model the user interests and is not amenable to modification.

Locally- vs. globally-fair PageRank. Existing papers [20, 29, 37] make the observation that if fairness is ensured at each step of the PageRank random walk, then fairness is guaranteed for the final PageRank distribution. Ensuring fairness at each step (which we call locally-fair PageRank) translates to having equal probability of visiting all groups from each vertex. While this is a sound approach, the requirement of locally-fair PageRank can be too rigid and may lead to a disproportionate modification of the graph structure or the edge weights. Instead, in this paper, we pose the question of *whether it is possible to achieve globally-fair PageRank without having to impose the strict requirement of locally-fair PageRank*.

As we will see, we answer this question in the affirmative.

Our approach. We present a new approach to incorporate fairness into the PageRank algorithm by reweighting the transition probabilities of the PageRank matrix. More concretely, we seek to minimize the PageRank distance to a fair distribution while allowing only for reweighting the network edges within prespecified limits. Our approach handles multiple vertex groups. Unlike earlier approaches, we do not recommend that new edges be added to the network, nor do we adjust the restart vector. Instead, we keep the topology of the underlying network unchanged.

If perfect fairness cannot be achieved, our formulation aims to find a solution as close as possible to the target fairness objective.

Our approach provides an actionable environment to improve PageRank fairness, as the newly computed edge weights can be lead to concrete actions, e.g., reweighting and prioritizing the feed of users in the social-network platform.

We propose a *projected gradient-descent* method for computing optimal edge reweightings and show how it can be implemented efficiently. We empirically compare our approach with state-of-the-art baselines and demonstrate the efficacy of our method, where small changes in the transition matrix lead to significant improvement in the fairness of the PageRank algorithm.

Contributions. Our main contributions are:

- We incorporate group fairness into PageRank by reweighting transition probabilities to achieve solutions as close as possible to the target fairness objective with minimal modifications.
- We aim for globally-fair PageRank without the strict requirement of achieving local fairness at each vertex. This can lead to solutions with smaller modifications in the transition matrix.
- We introduce a group-adapted fairness notion that accounts for group homophily via group-biased restart distributions.
- We propose an efficient projected gradient-descent method for finding locally-optimal edge weights.
- We empirically compare our approach with state-of-the-art baselines and demonstrate the efficacy of our method, where very small changes in the transition matrix lead to significant improvement in the fairness of the PageRank algorithm.

Due to space limitations, proofs and additional material are deferred to the appendices in the full version of this paper [38].

2 Related work

In recent years, algorithmic fairness in graph mining has emerged as a prominent area of research [32]. This paper focuses on enhancing fairness in the content of the well-known PageRank algorithm, which has been widely used for ranking, classification, community detection, and more. The objective is to allocate PageRank scores to vertices in a manner that considers both the vertices' group affiliations and the overall structure of the network [32, 37].

In the standard PageRank algorithm [6, 14, 34], a random-walk process defines the transition from one vertex to its neighbors, while also with certain probability considering a "jump" to one of the vertices defined by a restart vector. When considering the group association of vertices, defined by a protected attribute, analytical studies in the literature [11, 13] have unveild the unfairness of PageRank. To address such potential group bias, FairWalk [29] proposes to modify the transition matrix of the random walk so that the transition probabilities from each vertex to all groups become equal. CrossWalk [20] further refines this idea by considering the local group structure and reweighting edges based on their distance from group boundaries. Both FairWalk and CrossWalk adjust the transition matrix using heuristic-based methods.

Tsioutsiouliklis et al. [37] introduce a novel fairness metric for PageRank and propose two families of algorithms: FSPR and LFPRs. Methods in the first family, FSPR, retain the original graph structure and edge weights, while seeking an optimal jump vector to minimize the disparity between the fair PageRank vector and the original PageRank vector, subject to fairness constraints. On the other hand, similar to FairWalk and CrossWalk, methods in the second family, LFPRs, aim to achieve fairness at the individual vertex level and include three variants: LFPR_N, LFPR_U, and LFPR_P. Each variant reweights the transition matrix in a different way while incorporating a fair jump vector. Although the authors suggest that FSPR and LFPRs can be extended to scenarios involving more than two groups, they do not provide details for such an extension.

Additionally, link recommendation has been explored as an alternative approach to promoting fairness in PageRank. In one such paper, Tsioutsiouliklis et al. [36] present analytical formulas for calculating the impact of adding edges on fairness. In a different direction, Zeng et al. [43] propose a set of de-biasing methods for fair network representation learning, using techniques based on sampling, projections, and graph neural networks.

Other papers consider the problem of infusing fairness into rankings without focusing specifically into the PageRank algorithm. For instance, Krasanakis et al. [21] propose a model that weights training samples to make classifiers fair and adapt it to estimate an unbiased personalization that yields fairer vertex ranks.

In this paper, we propose an edge-reweighting approach to enhance fairness in PageRank. Our method differs from previous studies in several ways: First, we consider only edge reweighting, which can lead to more actionable operations to achieve fairness. Second, we focus on global fairness, without enforcing the more rigid requirement of fair transitions from each vertex. As a result, our method can lead to significantly smaller modification of the transition matrix. Finally, our approach offers a clear and practical framework for handling effectively fairness over multiple groups.

3 Problem definition

3.1 Preliminaries

We consider a directed graph $G = (V, E)$ with a set V of n vertices and a set E of m edges. We assume that the set of vertices is partitioned into K groups $\{V_1 \cdots V_K\}$. Let ℓ_v indicate the group that a vertex $v \in V$ belongs to.

The PageRank algorithm takes G as input, and produces a PageRank vector $\mathbf{p}$ that indicates the relative importance of each vertex in the graph [6]. Despite the existence of multiple equivalent formulations, the PageRank algorithm is fully characterized by three parameters: the transition matrix $\mathbf{P}$, the restart (or jump) vector $\mathbf{v}$, and the restart probability γ. The PageRank vector $\mathbf{p}$ is the solution to the equation $\mathbf{p}^\top = (1-\gamma)\mathbf{p}^\top\mathbf{P} + \gamma\mathbf{v}^\top$.

Here, the transition matrix $\mathbf{P}$ is row stochastic, derived by normalizing each row of the adjacency matrix of G. The restart vector $\mathbf{v}$ specifies a distribution over all vertices, indicating where a random walk might restart. The parameter γ represents the probability that the random walk will restart at any given step. In the case of directed graph, there may exist sink vertices that have no outgoing edges, so their rows in the transition matrix would be all zeros. To handle sink vertices, we replace the corresponding zero rows of $\mathbf{P}$ with the restart vector $\mathbf{v}$.

By solving the equation of PageRank definition, we obtain the PageRank vector as $\mathbf{p}^\top = \gamma\mathbf{v}^\top(\mathbf{I} - (1-\gamma)\mathbf{P})^{-1}$.

3.2 Fair PageRank

Building on the earlier definition of fairness-aware PageRank [36, 37], we consider the vertices of a graph to be partitioned into distinct groups. These groups may represent social categories based on sensitive attributes such as gender, ethnicity, or sexual orientation. We begin by defining the fair PageRank problem, followed by a discussion of various approaches to achieving PageRank fairness, and conclude with a formal definition of the $\boldsymbol{\phi}$-PageRank ($\boldsymbol{\phi}$-PageRank) problem and the Restricted $\boldsymbol{\phi}$-PageRank (R-ϕPR) problem.

3.2.1 Group-wise fair PageRank. To establish the concept of PageRank fairness, we assume a *target PageRank score*, ϕ_k for each vertex group $k = 1, \dots, K$, with $\sum_k \phi_k = 1$. This score represents the total PageRank mass that the algorithm is expected to allocate to the vertices in group V_k. A common approach is to set ϕ_k proportionally to the size of V_k, i.e., $\phi_k = |V_k|/n$. However, other choices may be more suitable depending on the specific application. In this work, as in prior studies, we assume that the target PageRank score ϕ_k for each group is provided as input to our problem formulation.

Let $\mathbf{1}_k \in \{0,1\}^n$ denote the indicator vector for group V_k, where $\mathbf{1}_k[i] = 1$ if vertex $i \in V_k$ and $\mathbf{1}_k[i] = 0$ otherwise. Let $\mathbf{p}$ be the PageRank vector computed by the PageRank algorithm using a transition matrix $\mathbf{P}$, with restart probability γ and restart vector $\mathbf{v}$. The total PageRank score for group V_k is then given by $\mathbf{1}_k^\top\mathbf{p}$.

Given an input graph G and its associated transition matrix $\mathbf{P}$, our goal is to find a new transition matrix $\hat{\mathbf{P}}$, defined on the same input graph G, such that the group-wise PageRank distribution computed over $\hat{\mathbf{P}}$ (defined shortly) closely matches the target group-wise PageRank distribution $\boldsymbol{\phi} = (\phi_1, \cdots, \phi_K)$.

Remark. For any given $\mathbf{P}$, γ, and $\mathbf{v}$, the total PageRank score for group V_k is lowered bounded by $\gamma\mathbf{1}_k^\top\mathbf{v}$ and upper bounded by $1-\gamma+\gamma\mathbf{1}_k^\top\mathbf{v}$. The lower bound represents the probability that a random walk restarts from group k, and it can be derived from the equation $\mathbf{1}_k^\top\mathbf{p} = (1-\gamma)\mathbf{1}_k^\top\mathbf{P}^\top\mathbf{p} + \gamma\mathbf{1}_k^\top\mathbf{v} \geq \gamma\mathbf{1}_k^\top\mathbf{v}$. The upper bound can be derived from the equation $1-\sum_{j\neq k}\mathbf{1}_j^\top\mathbf{v} \leq 1-\gamma\sum_{j\neq k}\mathbf{1}_j^\top\mathbf{v} = 1-\gamma+\gamma\mathbf{1}_k^\top$. Thus, for fixed γ and $\mathbf{v}$, it is clear that it may not always be possible to find a $\hat{\mathbf{P}}$ that exactly achieves the desired target distribution $\boldsymbol{\phi}$.

3.2.2 Rationale for PageRank fairness via edge reweighting. Consider the case where both $\mathbf{v}$ and γ can be freely adjusted. In this scenario, achieving the target fair distribution $\boldsymbol{\phi}$ is straightforward: we can simply set $\mathbf{v}[i] = \phi_{\ell_i}/|V_{\ell_i}|$ and $\gamma = 1$. However, adjusting $\mathbf{v}$ and γ is impractical as these parameters are not within our control in actual models; they are merely parameters of PageRank and do not reflect the actual graph structure. In contrast, modifying the transition matrix is more meaningful; this can be accomplished, for example, by appropriately prioritizing or demoting content from an individual's connections in a social network.

Therefore, we focus solely on adjusting the transition matrix $\mathbf{P}$ to achieve the desired fairness distribution. To maintain the structure of the input graph G, we impose the constraint that the new transition matrix $\hat{\mathbf{P}}$ must satisfy $\hat{\mathbf{P}}[i,j] = 0$ whenever $\mathbf{P}[i,j] = 0$. This condition ensures that no new edges are introduced.

3.2.3 Fairness-loss function and the $\boldsymbol{\phi}$-PageRank problem. To formalize our problem, we first define the fairness-loss function. It evaluates how much the actual fairness distribution deviates from the target distribution for a given transition matrix $\mathbf{P}$.

Definition 1 (PageRank Problem Instance)**.** *A* PageRank problem instance *consists of a graph G with vertices partitioned into groups $V_1, \dots, V_K$, a row-stochastic transition matrix $\mathbf{P}$, a restart probability $0 < \gamma < 1$, a restart vector $\mathbf{v} \in \mathbb{R}^n$ satisfying $\|\mathbf{v}\|_1 = 1$, and group-wise target PageRank scores $\boldsymbol{\phi} = (\phi_1, \dots, \phi_K)$ satisfying $\|\boldsymbol{\phi}\|_1 = 1$.*

Definition 2 (Fairness-loss function)**.** *Given a PageRank problem input as defined in Definition 1, the fairness-loss function is defined as:*

$$L(\mathbf{P}, \gamma, \mathbf{v}) = \frac{1}{K}\sum_{k=1}^{K}\left(\mathbf{1}_k^\top\mathbf{p} - \phi_k\right)^2, \tag{3.1}$$

where $\mathbf{p}$ is the PageRank vector corresponding to parameters $(\mathbf{P}, \gamma, \mathbf{v})$.

The proof of Proposition 1 can be found in Appendix A of the full version [38].

Proposition 1. *The fairness-loss function $L(\mathbf{P}, \gamma, \mathbf{v})$, as defined in Equation* (3.1)*, is not convex with respect to $\mathbf{P}$.*

We introduce two problems that aim to achieve fair PageRank by minimizing the fairness-loss through edge reweighting.

Problem 1 ($\boldsymbol{\phi}$-PageRank)**.** *Given a PageRank problem input as defined in Definition 1, the goal is to find a new transition matrix* $\mathbf{P}^*$ *that minimizes the fairness-loss function* $L(\mathbf{P}^*, \gamma, \mathbf{v})$*, i.e.,*

$$\mathbf{P}^* = \arg\min_{\hat{\mathbf{P}} \in \mathcal{C}(\mathbf{P})} L(\hat{\mathbf{P}}, \gamma, \mathbf{v}), \tag{3.2}$$

where

$$\mathcal{C}(\mathbf{P}) = \{\hat{\mathbf{P}} \in \mathbb{R}^{n\times n}_{\geq 0} \mid \hat{\mathbf{P}}\mathbf{1} = \mathbf{1},\ \hat{\mathbf{P}}_{ij} = 0 \text{ if } \mathbf{P}_{ij} = 0\}$$

is the set of row-stochastic matrices that have non-negative entries, and the edge reweighting of G can take place only on existing edges.

Problem 2 (Restricted $\boldsymbol{\phi}$-PageRank)**.** *Given a PageRank problem input as defined in Definition 1, a relative modification bound* $\delta \geq 0$ *and an absolute modification bound* $\epsilon \geq 0$*, the goal is to find a new transition matrix* $\mathbf{P}^*$ *that minimizes the fairness-loss function* $L(\mathbf{P}^*, \gamma, \mathbf{v})$*, i.e.,*

$$\mathbf{P}^* = \arg\min_{\hat{\mathbf{P}} \in \mathcal{C}_R(\mathbf{P}) \cap \mathcal{C}(\mathbf{P})} L(\hat{\mathbf{P}}, \gamma, \mathbf{v}), \tag{3.3}$$

where $\mathcal{C}(\mathbf{P})$ *is defined as in Problem 1, and*

$$\mathcal{C}_R(\mathbf{P}) = \{\hat{\mathbf{P}} \in \mathbb{R}^{n\times n}_{\geq 0} \mid (1-\delta)\mathbf{P}[i,j] - \epsilon \leq \hat{\mathbf{P}}[i,j] \leq (1+\delta)\mathbf{P}[i,j] + \epsilon\}.$$

In other words, $\mathcal{C}_R(\mathbf{P})$ restricts the entries of the new transition matrix $\hat{\mathbf{P}}$ within a range of $(1-\delta)$ and $(1+\delta)$ of the corresponding entries in the original transition matrix $\mathbf{P}$, while also imposing an absolute modification bound ϵ. The relative bound is necessary because, for very small values in $\mathbf{P}$, even slight absolute changes can result in large relative variations. By choosing δ and ϵ such that $(1-\delta)\mathbf{P} - \epsilon > 0$, we can guarantee no edges are deleted.

The motivation for defining the Restricted $\boldsymbol{\phi}$-PageRank problem (Problem 2) is to consider only solutions that make small modifications in the transition matrix $\mathbf{P}$.

Without loss of generality, in this paper, we study both $\boldsymbol{\phi}$-PageRank and Restricted $\boldsymbol{\phi}$-PageRank with a uniform restart vector $\mathbf{v} = \left[\frac{1}{n}, \dots, \frac{1}{n}\right]$.

3.3 Group-adapted fair PageRank

Real-world networks are mostly homophilic, i.e., vertices in the same group are more likely to be connected to each other [24, 35]. Hence, the total PageRank scores in group V_k are largely contributed by random walks (re)starting at vertices in group V_k [37]. Thus, we propose the notion of *group-adapted fairness*, which enhances PageRank by considering random walks that (re)start at each group and by penalizing the fairness loss for each such (re)start.

More specifically, let $\mathbf{v}_k = \frac{1}{|V_k|}\mathbf{1}_k$ be the vector indicating that the random walk restarts uniformly only from group k. We define the *group-adapted fairness-loss function* as follows:

Definition 3 (Group-adapted fairness loss)**.** *Given a PageRank problem input as defined in Definition 1, the group-adapted fairness-loss function is defined as*

$$L_g(\mathbf{P}, \gamma) = \frac{1}{K}\sum_{\ell=1}^{K} L(\mathbf{P}, \gamma, \mathbf{v}_\ell) = \frac{1}{K^2}\sum_{\ell=1}^{K}\sum_{k=1}^{K}\left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right)^2, \tag{3.4}$$

where $\mathbf{p}_\ell$ *is the PageRank vector with respect to parameters* $(\mathbf{P}, \gamma, \mathbf{v}_\ell)$*.*

In the group-adapted fairness loss, we iterate through all groups and sum up the fairness loss when restarting the random walk at each group. Minimizing the group-adapted fairness-loss function leads to the Group-adapted $\boldsymbol{\phi}$-PageRank problem.

Problem 3 (Group-adapted $\boldsymbol{\phi}$-PageRank)**.** *Given a PageRank problem input as defined in Definition 1, find a new transition matrix* $\mathbf{P}^*$ *that minimizes the group-adapted fairness-loss function*

$$\mathbf{P}^* = \arg\min_{\hat{\mathbf{P}} \in \mathcal{C}(\mathbf{P})} L_g(\hat{\mathbf{P}}, \gamma), \tag{3.5}$$

where $\mathcal{C}(\mathbf{P})$ *is defined as in Problem 1.*

Similarly, we define the restricted version of the problem.

Problem 4 (Restricted group-adapted $\boldsymbol{\phi}$-PageRank)**.** *Given a PageRank problem input as defined in Definition 1, a relative modification bound* $\delta \geq 0$ *and an absolute modification bound* $\epsilon \geq 0$*, find a new transition matrix* $\mathbf{P}^*$ *that minimizes the group-adapted fairness-loss function* $L_g(\mathbf{P}^*, \gamma)$*, i.e.,*

$$\mathbf{P}^* = \arg\min_{\hat{\mathbf{P}} \in \mathcal{C}_R(\mathbf{P}) \cap \mathcal{C}(\mathbf{P})} L_g(\hat{\mathbf{P}}, \gamma),$$

where $\mathcal{C}(\mathbf{P})$ *is defined as in Problem 1 and* $\mathcal{C}_R(\mathbf{P})$ *as in Problem 2.*

4 Algorithms

In this section, we present a projected gradient-descent algorithm for finding a locally-optimal transition matrix $\hat{\mathbf{P}}$ for the fair PageRank problems we defined in the previous section. We propose Algorithm 1 for problem Problem 1 and Problem 2, and Algorithm 3 for problem Problem 3 and Problem 4.

4.1 Projected gradient descent for ϕ-PageRank

We first derive the gradient of the fairness-loss function $L(\mathbf{P}, \gamma, \mathbf{v})$ with respect to the transition matrix $\mathbf{P}$. The proof of the following proposition can be found in Appendix B of the full version [38].

Proposition 2. *The gradient of the fairness-loss function* $L(\mathbf{P}, \gamma, \mathbf{v})$ *with respect to the transition matrix* $\mathbf{P}$ *is given by*

$$\frac{\partial}{\partial \mathbf{P}} L(\mathbf{P}, \gamma, \mathbf{v}) = \frac{2(1-\gamma)}{K}\sum_{k=1}^{K}\left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right)\mathbf{p}\mathbf{y}_k^\top, \tag{4.1}$$

where $\mathbf{p}$ *is the PageRank vector with respect to* $(\mathbf{P}, \gamma, \mathbf{v})$*, and* $\mathbf{y}_k = \mathbf{U}^{-T}\mathbf{1}_k = (\mathbf{I} - (1-\gamma)\mathbf{P})^{-1}\mathbf{1}_k$*, with* $\mathbf{U} = \mathbf{I} - (1-\gamma)\mathbf{P}^\top$*.*

Matrix inversion acceleration. The calculation of $\mathbf{y}_k = \gamma\mathbf{U}^{-T}\mathbf{1}_k$ involves the inverse of the matrix $\mathbf{U}^\top = \mathbf{I} - (1-\gamma)\mathbf{P}$, which is computationally expensive. To avoid computing matrix inversion, we approximation $\mathbf{U}^{-T}$ with the Neumann series $\mathbf{U}^{-T} = \sum_{i=0}^{\infty}(1-\gamma)^i\mathbf{P}^i$, and substituting into $\mathbf{y}_k$ gives $\mathbf{y}_k = \sum_{i=0}^{\infty}(1-\gamma)^i\mathbf{P}^i\mathbf{1}_k$. As the series terms decrease exponentially in magnitude, due to the decaying factor $(1-\gamma)$, we can approximate $\mathbf{y}_k$ efficiently by computing only the first few series terms. We show that truncating the Neumann series to its first 50 terms yields a relative error below 0.0003. The detailed analysis is provided in Appendix D of the full version [38].

Gradient interpretation. In Proposition 2 we have defined $\mathbf{y}_k = \mathbf{U}^{-T}\mathbf{1}_k$. The j-th coordinate of $\mathbf{y}_k$ is given by $\mathbf{y}_k[j] = \mathbf{1}_k^\top(\gamma\mathbf{U}^{-1}\mathbf{e}_j)$, which represents the sum of PageRank scores for the vertices in the

Algorithm 1 `FairGD`: Projected gradient descent for $\boldsymbol{\phi}$-PageRank and Restricted $\boldsymbol{\phi}$-PageRank

Input: $\mathbf{P}$, γ, $\mathbf{v}$, $\boldsymbol{\phi} = (\phi_1, \dots, \phi_K)$, group indicator vectors $\mathbf{1}_k$ for $k = \{1, \cdots, K\}$, learning rate α, iteration exponent t_1 and t_2, convergence criterion κ, maximum number of iterations N_{iter}, modification bound δ and ϵ (optional).
Output: Optimized transition matrix $\hat{\mathbf{P}}$

$\hat{\mathbf{P}} \leftarrow \mathbf{P}$, $iter \leftarrow 0$, $L_{prev} \leftarrow \infty$, $\mathbf{p} = \frac{1}{n}\mathbf{1}$
while $|L(\hat{\mathbf{P}}, \gamma, \mathbf{v}) - L_{prev}| > \kappa$ and $iter < N_{\text{iter}}$ **do**
  $iter \leftarrow iter + 1$
  $\mathbf{p} \leftarrow$ iterating $\mathbf{p}^\top = (1-\gamma)\mathbf{p}^\top\hat{\mathbf{P}} + \gamma\mathbf{v}^\top$ for t_1 times
  $L_{prev} \leftarrow L(\hat{\mathbf{P}}, \gamma, \mathbf{v})$
  for $k = 1$ to K **do**
    $\mathbf{y}_k \leftarrow \sum_{i=1}^{t_2}(1-\gamma)^i\hat{\mathbf{P}}^i\mathbf{1}_k$
    $\hat{\mathbf{P}} \leftarrow \hat{\mathbf{P}} - \alpha\frac{2(1-\gamma)}{K}(\mathbf{1}_k^\top\mathbf{p} - \phi_k)\mathbf{p}\mathbf{y}_k^\top$
  end for
  $\hat{\mathbf{P}} \leftarrow$ `Project`$(\hat{\mathbf{P}}, \delta, \epsilon)$
end while
return $\hat{\mathbf{P}}$

Algorithm 2 `Project`: projection to feasible set

Input: Transition matrix $\hat{\mathbf{P}}$, modification bound δ and ϵ (optional).
Output: Projected transition matrix $\hat{\mathbf{P}}$

1 **for** $i = 1, \cdots, n$ **do**
2 $\mathbf{s} \leftarrow$ nonzero elements of $\hat{\mathbf{P}}_i$
3 **if** δ and ϵ are provided **then**
4 $\boldsymbol{\ell} = \max(0, \mathbf{s}(1-\delta) - \epsilon)$
5 $\mathbf{u} = \min(1, \mathbf{s}(1+\delta) + \epsilon)$
6 $\lambda^* \leftarrow$ `Root-Finding`$(\mathbf{s}, \boldsymbol{\ell}, \mathbf{u})$
7 $s_j^* = \text{clip}(\lambda^* + s_j, \ell_j, u_j)$ for all $j \in \{1, \cdots, |\mathbf{s}|\}$
8 **else**
9 Rank elements in $\mathbf{s}$ such that $s_1 \geq \cdots \geq s_{|\mathbf{s}|}$
10 $k(\mathbf{s}) = \max\left\{k \in \{1, \dots, |\mathbf{s}|\} \mid 1 + k\, s_k > \sum_{j \leq k} s_j\right\}$
11 $\tau(\mathbf{s}) = \frac{1}{k(\mathbf{s})}\left(\sum_{j \leq k(\mathbf{s})} s_j - 1\right)$
12 $s_j^* = \max(0, s_j - \tau(\mathbf{s}))$ for all $j \in \{1, \cdots, |\mathbf{s}|\}$
13 **end if**
14 non-zeros entries of $\hat{\mathbf{P}}_i \leftarrow \mathbf{s}^*$
15 **end for**

group V_k, with the random walk restarting at the unit vector $\mathbf{e}_j$. In other words, the random walk always (re)starts from vertex j.

Hence, the gradient of the fairness loss has the following intuitive interpretation: the $[i, j]$-th entry of the gradient matrix is a weighted sum of the products of $\mathbf{p}[i]$ and $\mathbf{y}_k[j]$. The coordinate $\mathbf{p}[i]$ represents the frequency with which a random walk, restarting at $\mathbf{v}$, visits vertex i, while the coordinate $\mathbf{y}_k[j]$ represents the frequency with which a random walk, restarting at vertex j, visits vertices in the group V_k. Therefore, the k-th term of the summation in Eq. (4.1) reflects the weighted contribution of the edge (i, j) to the overall PageRank score of the group V_k. The weight of the k-th term in Eq. (4.1) is proportional to the difference between the total PageRank scores of group V_k and the group-wise target score ϕ_k.

Gradient projection. After each gradient descent step, $\hat{\mathbf{P}}$ needs to be projected back into the feasible area subject to constraints $\mathcal{C}(\mathbf{P})$ or $\mathcal{C}(\mathbf{P}) \cap \mathcal{C}_R(\mathbf{P})$. We perform the projection for each row of $\hat{\mathbf{P}}$ separately, and since both constraints forbid adding non-existing edges, we solve the projection only for its non-zero entries. Given any row of $\hat{\mathbf{P}}$, let $\mathbf{s}$ be the vector comprises the non-zero entries of the corresponding row vector, and $\mathbf{s}^*$ be the vector after projection. We use $N = |\mathbf{s}|$ to denote the cardinality of $\mathbf{s}$.

We first discuss the constraint $\mathcal{C}(\mathbf{P})$. Projecting $\mathbf{s}$ onto a vector $\mathbf{t}$ that lies in the feasible set $\mathcal{C}(\mathbf{P})$ is equivalent to solving the following problem:

$$\mathbf{s}^* = \arg\min_{\mathbf{t} \in \mathcal{F}_1} \|\mathbf{s} - \mathbf{t}\|_2^2,$$

where $\mathcal{F}_1 = \{\mathbf{t} \in \mathbb{R}^N \mid \mathbf{1}^\top\mathbf{t} = 1, \mathbf{t} \geq 0\}$ is a probability simplex. This is a well studied problem [9, 10, 25, 26], and the computational complexity for finding $\mathbf{s}^*$ is $\mathcal{O}(NlogN)$. In Algorithm 2, lines 9–12, we use the method detailed by Duchi et al. [10] to perform the projection of $\mathbf{s}$ after every gradient update step.

We then discuss the constraint $\mathcal{C}(\mathbf{P}) \cap \mathcal{C}_R(\mathbf{P})$. Solving the projection subject to this constraint is equivalent to finding a Euclidean projection of $\mathbf{s}$ onto the intersection of a probability simplex and a box. Namely,

$$\mathbf{s}^* = \arg\min_{\mathbf{t} \in \mathcal{F}_2} \|\mathbf{s} - \mathbf{t}\|_2^2,$$

where $\mathcal{F}_2 = \{\mathbf{t} \in \mathbb{R}^N \mid \mathbf{1}^\top\mathbf{t} = 1, \ell_i \leq t_i \leq u_i, \text{ for all } i \in \{1, \cdots, N\}\}$, with $\ell_i = \max\{0, (1-\delta)s_i - \epsilon\}$ and $u_i = \min\{1, (1+\delta)s_i + \epsilon\}$. This problem has already been studied by Adam and Mácha [1], and the time complexity for finding $\mathbf{s}^*$ is also $\mathcal{O}(NlogN)$. We provide the details in Algorithm 2, lines 4–7. Line 6 invokes Algorithm 4 (`Root-Finding`) as a subroutine, and the details are provided in Appendix E of the full version [38].

To sum up, our method is presented in Algorithms 1 and 2. Algorithm 1 (`FairGD`) is the overall projected gradient-descent scheme that iteratively updates the matrix $\hat{\mathbf{P}}$ until convergence. Algorithm 2 (`Project`) is the projection function, where the transition matrix $\hat{\mathbf{P}}$ is projected back to the feasible set after each iteration.

Algorithm 1 is designed to address both the $\boldsymbol{\phi}$-PageRank problem and the Restricted $\boldsymbol{\phi}$-PageRank problem. In the non-restricted case, we refer to the algorithm simply as `FairGD`. In the restricted case, parameterized by (δ, ϵ), we denote it as `FairGD`(δ, ϵ).

4.2 Group-adapted PageRank

The `FairGD` algorithm can be extended to address both the Group-adapted $\boldsymbol{\phi}$-PageRank and Restricted group-adapted $\boldsymbol{\phi}$-PageRank problems by introducing an outer loop that iterates over all groups. This enhanced algorithm, referred to as `AdaptGD`, is detailed in Algorithm 3 in Appendix E of the full version [38]. For the non-restricted case, we refer to the algorithm as `AdaptGD`. In the restricted case, parameterized by (δ, ϵ), we denote it as `AdaptGD`(δ, ϵ).

Proposition 3. *The gradient of the fairness-loss function $L_g(\mathbf{P}, \gamma)$ with respect to the transition matrix $\mathbf{P}$ is given by*

$$\frac{\partial}{\partial \mathbf{P}} L_g(\mathbf{P}, \gamma) = \frac{2(1-\gamma)}{K^2} \sum_{k=1}^{K} \sum_{\ell=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right)\left(\mathbf{p}_\ell \mathbf{1}_k^\top \mathbf{U}^{-1}\right). \quad (4.2)$$

where $\mathbf{p}_\ell$ is the PageRank vector with respect to $(\mathbf{P}, \gamma, \mathbf{v}_\ell)$.

4.3 Extensions

While not explicitly studied in the paper, we note that the proposed fairness loss can be generalized in several ways. First, when fairness scores are specified at the individual-node level, an auxiliary loss term can jointly enforce both group-level and individual-level constraints. Second, for graphs with hierarchical group structures, where each layer defines a partition with associated fairness goals, the overall objective can be formulated as the sum of layer-specific losses, enabling simultaneous satisfaction of fairness requirements across multiple scales. Finally, the restricted fairness losses can incorporate node-level constraints by specifying δ and ϵ for each node, with our algorithm remaining applicable.

5 Complexity and convergence analysis

The primary computational cost arises from computing PageRank and approximating $\mathbf{y}_k$. The PageRank vector is obtained by iterating the update equation $\mathbf{p}^\top = (1-\gamma)\,\mathbf{p}^\top\mathbf{P} + \gamma\,\mathbf{v}^\top$ for t_1 iterations, resulting in computational complexity $\mathcal{O}(t_1 n^2)$. The computation of $\mathbf{y}_k$ involves multiplying a vector by a matrix t_2 times. Evaluating $\mathbf{y}_k$ for all $k \in \{1, \dots, K\}$ therefore requires $\mathcal{O}(K t_2 n^2)$ operations. Consequently, the overall complexity of Algorithm 1 is $\mathcal{O}\big((K t_2 + t_1) n^2\big)$. Since AdaptGD performs an additional loop over K groups when computing $\mathbf{y}_k$, its complexity becomes $\mathcal{O}\big((K^2 t_2 + t_1) n^2\big)$.

Both loss objective functions, $L(\mathbf{P}, \gamma, \mathbf{v})$ and $L_g(\mathbf{P}, \gamma)$, are differentiable functions concerning $\mathbf{P}$, and their first derivatives are smooth. Since the feasible set $\mathcal{C}_R(\mathbf{P}) \cap \mathcal{C}(\mathbf{P})$ is convex and closed, the proposed projected gradient update method with constant step size converges to a stationary point, which is a local minimum [5]. We provide detailed analysis in Appendix F and Appendix G of the full version.

6 Experimental evaluation

In our experimental evaluation, we study the following questions:

Q1: To what extent can the FairGD and AdaptGD algorithms improve the fairness and group-adapted fairness of PageRank?

Q2: How do the FairGD and AdaptGD algorithms compare to the baselines in terms of changes to the transition matrix?

Q3: How does the ranking of the PageRank score within each group change? Do FairGD and AdaptGD preserve this ranking?

To answer the first question, we report the fairness loss and group-adapted fairness loss as defined in Equations (3.1) and (3.4). For the second question, we quantify the changes to the transition matrix by the relative change, given by $\Delta_{\mathbf{P}} = \|\hat{\mathbf{P}} - \mathbf{P}\|_F / \|\mathbf{P}\|_F$. For the third question, we compute the Spearman's rank correlation coefficient [33] between the original PageRank ranking and the ranking after re-weighting, and report the weighted average $\overline{\rho}$ across all groups (weighted by the group size). A larger $\overline{\rho}$ indicates fewer changes in the PageRank score's ranking.

6.1 Datasets

We evaluate our methods and the baselines on the following publicly-available datasets: Books,[1] Blogs [2], MIND [40], Twitter [31], and Slashdot [22]. Dataset statistics are given in Table 1, whereas more information is presented in Appendix H.1 of the full version [38].

[1] http://www-personal.umich.edu/~mejn/netdata/

Table 1: Datasets used in the experiments. $\boldsymbol{\phi}_O$ denotes the original group-wise PageRank scores of the datasets.

Dataset	#Vertices	#Edges	Modularity	$\boldsymbol{\phi}_O$
Books	105	882	0.4149	(0.42,0.10, 0.48)
Blogs	1224	19025	0.4111	(0.48, 0.52)
MIND	2269	11345	0.075	(0.26,0.74)
Twitter	18470	48365	0.4755	(0.42, 0.58)
Slashdot	82168	948464	0.2950	(0.30, 0.36, 0.17,0.17)

6.2 Proposed methods and baselines

The proposed methods are FairGD and AdaptGD. Additionally:

FairGD(δ, ϵ) is the restricted version of FairGD, with relative and absolute intervention bound δ and ϵ respectively.

AdaptGD(δ, ϵ) is the restricted version of our AdaptGD, with relative and absolute intervention bound δ and ϵ respectively.

We compare our methods with the following baselines: LFPR_U [37], LFPR_N [37], FairWalk [29], and CrossWalk [20]. More details on the baselines can be found in H.2 of the full version [38].

Remarks. Originally, FairWalk and CrossWalk assign equal weights to each group in the random walk. We modify them so that the probability of visiting each group during a random walk is proportional to the target group-wise PageRank scores. These two algorithms do not add edges, neither do they change the restart vector.

LFPR_N and LFPR_U are guaranteed to achieve perfect $\boldsymbol{\phi}$-fairness, provided the restart vector is $\boldsymbol{\phi}$-fair, i.e., $\mathbf{1}_k^\mathsf{T}\mathbf{v} = \phi_k$. LFPR_U modifies the graph to be fully connected. We report fairness loss $L(\hat{\mathbf{P}}, \gamma, \mathbf{v})$ and group-adapted fairness loss $L_g(\hat{\mathbf{P}}, \gamma)$ for these two baselines with the restart vector $\mathbf{v}$ set to a uniform vector.

6.3 Experimental setting

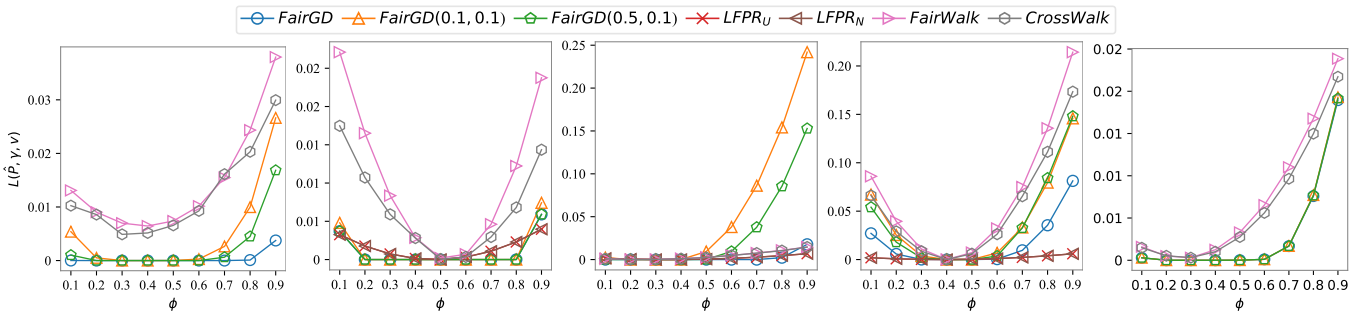
Parameters γ and $\mathbf{v}$. We follow the common assumption of PageRank algorithm and fix $\gamma = 0.15$. We fix $\mathbf{v}$ to be the uniform vector.

Group indicator definition. For datasets Books, Blogs, MIND, and Twitter, we use node labels as group indicators. For the Slashdot dataset, where node labels are unavailable, we apply the METIS graph partitioning algorithm [17] to divide the vertices into groups.

(Group-adapted) fairness loss evaluation. For the two-group datasets Blogs, MIND and Twitter datasets, we compare our approach with all baselines. For the remaining multi-group datasets, we only compare with FairWalk and CrossWalk as LFPR_U and LFPR_N do not extend directly to the multi-group scenario.

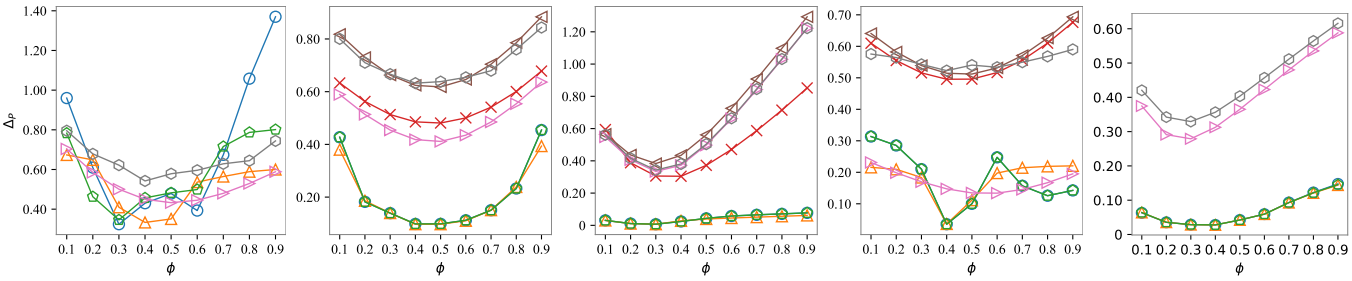
FairGD(δ, ϵ) and AdaptGD(δ, ϵ). For Group-adapted $\boldsymbol{\phi}$-PageRank and Restricted group-adapted $\boldsymbol{\phi}$-PageRank problems, we set $\delta = \{0.1, 0.5\}$ and $\epsilon = 0.1$. We perform a grid search over a logarithmically-spaced range of values between 10^{-4} and 10^4 to find the best learning rate.

Group-wise target PageRank score $\boldsymbol{\phi}$. For datasets containing only two groups, we define the target PageRank distribution as



(a) Books (b) Blogs (c) MIND (d) Twitter (e) Slashdot

Figure 1: Fairness loss $L(\hat{\mathbf{P}}, \gamma, \mathbf{v})$ of each method on five datasets for the $\boldsymbol{\phi}$-PageRank problem



(a) Books (b) Blogs (c) MIND (d) Twitter (e) Slashdot

Figure 2: Transition matrix change $\Delta_{\mathbf{P}}$ of each method on five datasets for the $\boldsymbol{\phi}$-PageRank problem

$\boldsymbol{\phi} = (\phi, 1-\phi)$. For datasets with multiple groups, we set $\boldsymbol{\phi} = \left(\phi, \frac{1-\phi}{K-1}, \cdots, \frac{1-\phi}{K-1}\right)$. We vary ϕ in the range $[0.1, 0.2, \ldots, 0.9]$.

The code and datasets used in this paper are available.[2]

6.4 Fairness loss and weight change $\Delta_{\mathbf{P}}$

We evaluate the fairness loss $L(\hat{\mathbf{P}}, \gamma, \mathbf{v})$ on `FairGD`, `FairGD`(0.1, 0.1), `FairGD`(0.5, 0.1), and all baselines across the five datasets; the results are shown in Figure 1. We also evaluate the changes, denoted as $\Delta_{\mathbf{P}}$, to the original transition matrix; the results are shown in Figure 2. The results demonstrate that `FairGD` and its restricted variants generally achieve better or comparable fairness loss compared to the baselines, while maintaining significantly smaller changes in the transition matrix weights.

On the two-group datasets, $\mathsf{LFPR}_{\mathsf{U}}$ and $\mathsf{LFPR}_{\mathsf{N}}$ achieve near-perfect fairness by modifying the graph so that each node fairly distributes its PageRank score to its neighbors according to the target fairness vector. On the Blogs dataset, `FairGD` and its two restricted variants exhibit very similar behavior, achieving 8 out of the 10 best fairness outcomes while also minimizing weight changes. For the Twitter dataset, `FairGD` and its variants outperform FairWalk and CrossWalk in terms of fairness loss, whereas FairWalk performs comparably with our proposed methods in weight change, surpassing all other baselines. On the MIND dataset, `FairGD` and all baselines achieve almost perfect fairness, but `FairGD` consistently outperforms them in weight change. However, for $\phi > 0.5$, the constrained variants `FairGD`(0.1, 0.1) and `FairGD`(0.5, 0.1) fail to maintain high fairness but consistently achieve smaller weight changes.

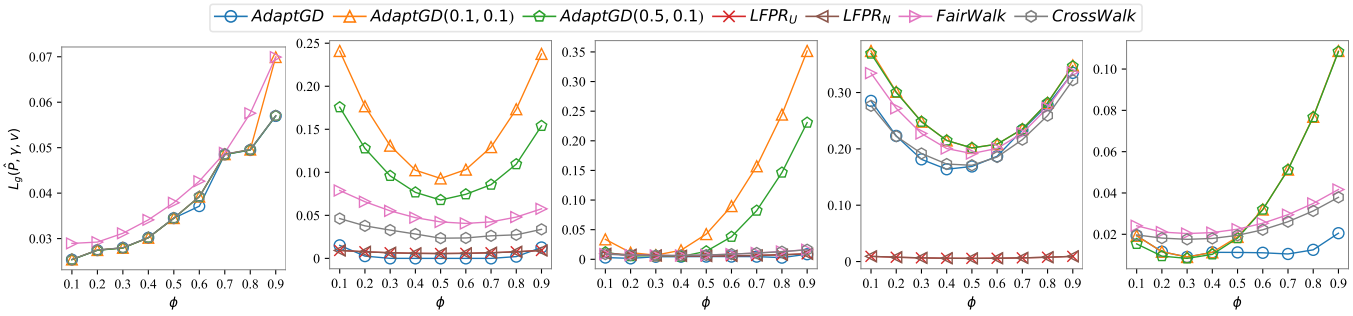
For multi-group datasets, `FairGD` and its two variants perform best on both fairness and weight change on the Slashdot dataset. On the Books dataset, `FairGD` attains the highest fairness score, followed by `FairGD`(0.5, 0.1) and `FairGD`(0.1, 0.1); the weight changes surpass those of FairWalk and CrossWalk for some ϕ values to achieve better fairness outcomes.

Overall, `FairGD` delivers superior results in terms of fairness loss and weight change when ϕ is close to the original PageRank score, ϕ_1 (the first dimension of $\boldsymbol{\phi}_o$ as reported in Table 1). As ϕ diverges from ϕ_1, both the fairness loss and weight change increase. The two restricted variants, `FairGD`(0.1, 0.1) and `FairGD`(0.5, 0.1), exemplify the trade-off between fairness loss and weight change by achieving smaller weight changes at the cost of higher fairness loss compared to `FairGD`. This refinement highlights the balance between fairness and weight change, emphasizing the flexibility of `FairGD` in adapting to various fairness goals.

6.5 Group-adapted fairness loss and $\Delta_{\mathbf{P}}$

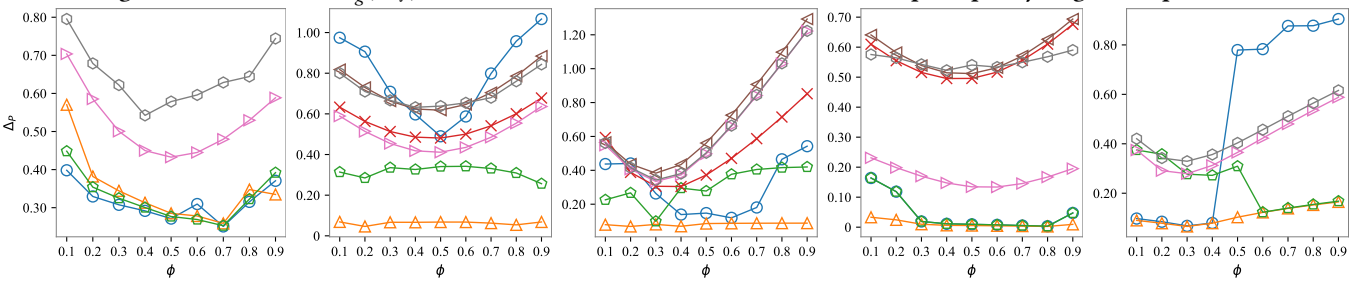
We evaluate `AdaptGD` and its restricted variants, `AdaptGD`(0.1, 0.1) and `AdaptGD`(0.5, 0.1), on the group-adapted fairness loss, as defined in Equation (3.4), and analyze the corresponding changes in the transition matrix weights. The results are presented in Figure 3

[2]https://github.com/HongLWang/Fair-Pagerank-via-Edge-Rewighting



(a) Books (b) Blogs (c) MIND (d) Twitter (e) Slashdot

Figure 3: Fairness loss $L_g(\hat{\mathbf{P}}, \gamma)$ of each baseline on five datasets for the Group-adapted ϕ-PageRank problem



(a) Books (b) Blogs (c) MIND (d) Twitter (e) Slashdot

Figure 4: Transition matrix change $\Delta_{\mathbf{P}}$ of each baseline on five datasets for the Group-adapted ϕ-PageRank problem

Table 2: Average Spearman's coefficient $\overline{\rho}$ between the original and revised PageRank weight for the MIND dataset

ϕ	FairWalk	CrossWalk	LFPR_U	LFPR_N	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	0.811	0.812	0.790	0.802	**0.970**	0.953	**0.970**	0.817	0.853	0.830
0.2	0.830	0.823	0.808	0.801	**0.997**	**0.997**	**0.997**	0.810	0.855	0.817
0.3	0.818	0.808	0.817	0.810	**0.998**	**0.998**	**0.998**	0.797	0.835	0.815
0.4	0.828	0.795	0.811	0.795	**0.970**	0.966	**0.970**	0.809	0.847	0.813
0.5	0.819	0.812	0.810	0.802	**0.970**	0.909	0.960	0.814	0.828	0.802
0.6	0.821	0.809	0.788	0.807	**0.966**	0.888	0.903	0.814	0.846	0.825
0.7	0.816	0.805	0.789	0.786	**0.958**	0.884	0.888	0.816	0.836	0.810
0.8	0.794	0.805	0.794	0.791	**0.930**	0.872	0.883	0.769	0.849	0.791
0.9	0.810	0.800	0.802	0.798	**0.884**	0.857	0.867	0.760	0.833	0.801

for the group-adapted fairness loss and in Figure 4 for the transition matrix weight changes. The results show that the trade-off between fairness loss and weight change is further exemplified and shown for AdaptGD and its variants.

Across the Books, Blogs, and MIND datasets, AdaptGD consistently achieves the most favorable fairness outcomes compared to competing baselines. In terms of weight change, AdaptGD attains optimal performance on the Books dataset and 9 out of 10 optimal outcomes on the MIND dataset. Although AdaptGD underperforms LFPR_U and LFPR_N in fairness loss on the Twitter dataset, it still matches or surpasses the other two baseline methods, while delivering substantially better weight change results. On the Slashdot dataset, AdaptGD outperforms all competing baselines in fairness outcome, and achieves the best weight change metrics for 4 out of 10 values of ϕ. Notably, on the Blogs and the Slashdot dataset, AdaptGD prioritizes improvements in fairness over weight change performance.

For all datasets, the two variants of AdaptGD exhibit comparable or slightly higher fairness loss relative to AdaptGD, while consistently maintaining lower or comparable weight change.

6.6 Ranking coefficient evaluation

We report the average Spearman's rank correlation coefficient $\overline{\rho}$ for each method across multiple datasets. For each group, we compute the Spearman's coefficient between the rankings of old and new PageRank values restricted to that group. These per-group coefficients are then aggregated using a weighted average, where each group's weight corresponds to its proportion of the total items. The results for the MIND dataset are shown in Table 2, while results for other datasets are provided in Appendix I of the full version [38]. In the tables, we depict with boldface the largest value and with underline the second largest.

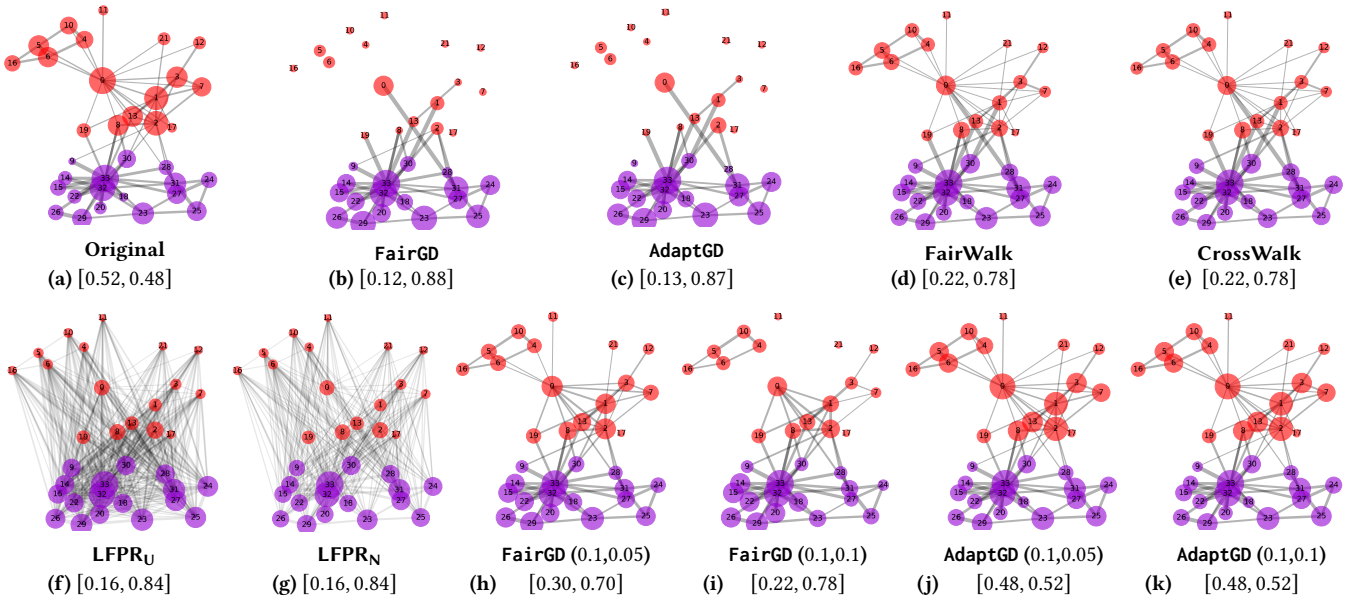


Figure 5: Graph structure of Karate before and after edge reweighting by each method. The target group-wise fairness vector is $[0.1, 0.9]$. Below each method name, the group-wise PageRank scores calculated using a uniform restart vector are displayed. The two values correspond to the red and purple groups, respectively.

Across all ϕ values, FairGD consistently attains the highest $\overline{\rho}$, followed by FairGD(0.5, 0.1) and FairGD(0.1, 0.1). These results indicate that FairGD exhibits very good preservation of the relative PageRank score rankings within individual groups. These results suggest that, in practical applications, when adjusting the content displayed to users based on items' PageRank scores to enhance fairness, fewer changes are required within each group. Instead, greater focus should be placed on adjusting the frequency or weighting of content across different groups.

6.7 Use case

We visualize the graph structure of a small real-world dataset, Karate [30, 41], before and after edge reweighting in Figure 5. The Karate dataset is a classic social network from Wayne Zachary's 1977 study of friendships among 34 members of a karate club, which eventually split into two factions after a disagreement. The dataset contains 34 nodes, with 17 nodes belonging to the "Mr. Hi" club and the remaining 17 nodes belonging to the "Officer" club.

In the visualization, vertex labels indicate node indices, node colors correspond to group membership, node sizes are proportional to their PageRank scores, and edge thickness reflects transition weights.

The original group-wise PageRank score for the Karate dataset is $[0.52, 0.48]$, while the fairness goal is set to $[0.1, 0.9]$. From Figure 5, we observe that FairGD achieves the closest group-wise fairness scores, followed by AdaptGD, LFPR$_U$, and LFPR$_N$. The unrestricted versions of FairGD and AdaptGD tend to remove a large proportion of intra-group edges within the red group, whereas their restricted counterparts delete fewer edges due to the modification bounds, at the cost of achieving a lower fairness improvement. FairWalk and CrossWalk remove no edges, resulting in graph structures very similar to the original. Finally, LFPR$_U$ modifies the graph to be fully connected, achieving the same fairness score as LFPR$_N$, which adds fewer non-existing edges.

7 Conclusion

We presented a novel approach for incorporating group fairness into the PageRank algorithm by reweighting the entries of the underlying transition matrix. We focus on matrix reweighting interventions, as we argue that those are more actionable strategies, and further, reflect the network structure, rather than the hyperparameters of PageRank. Our goal is to minimize fairness loss (with respect to a target group-fairness distribution) while also minimizing the changes to the transition matrix. We achieve our objective by foregoing the idea of ensuring fairness locally, at every vertex, and instead aim to ensure fairness at a global level. Technically, our method is an efficient projected gradient-descent algorithm for finding locally-optimal edge weights. We presented a comprehensive empirical evaluation, demonstrating that our method offers significant advantages compared to strong baselines.

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8 Ethical considerations

This work focuses on reweighting the transition matrix of a graph to achieve target group-wise PageRank scores, with the goal of improving fairness across groups. While the proposed approach aims to improve fairness and mitigate bias in ranking systems, interventions that alter item rankings could be misused to deliberately amplify or suppress the visibility of certain content, potentially leading to manipulation or censorship. Furthermore, if applied to sensitive domains, such interventions could unintentionally disadvantage specific users or groups, even when fairness metrics improve. Our experiments use only publicly available datasets with no personal user or item-identifying information, reducing privacy risks. To mitigate potential misuse, we recommend deploying such methods with transparent governance, independent auditing of fairness outcomes, and clear documentation of intervention goals and criteria.

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A Proof of Proposition 1

To demonstrate the non-convexity, it suffices to provide a single counterexample. Consider a fully connected graph G with three vertices: a, b, and c. Let the labels be ℓ_a = blue, and $\ell_b = \ell_c$ = red.

We also let

$$\hat{\mathbf{P}}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \text{ and } \hat{\mathbf{P}}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

be two row-stochastic transition matrices, and

$$\hat{\mathbf{P}}_3 = \frac{1}{2}\hat{\mathbf{P}}_1 + \frac{1}{2}\hat{\mathbf{P}}_2 = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}.$$

Given restart vector $\mathbf{v} = (1/3, 1/3, 1/3)$ and restart probability γ, we can calculate the PageRank vectors $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$, for $\hat{\mathbf{P}}_1$, $\hat{\mathbf{P}}_2$, and $\hat{\mathbf{P}}_3$, being $\mathbf{p}_1 = \left(\frac{\gamma^2-3\gamma+3}{3(2-\gamma)}, \frac{3-2\gamma}{3(2-\gamma)}, \frac{\gamma}{3}\right)$, $\mathbf{p}_2 = \left(\frac{\gamma^2-3\gamma+3}{3(2-\gamma)}, \frac{\gamma}{3}, \frac{3-2\gamma}{3(2-\gamma)},\right)$, and $\mathbf{p}_3 = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, respectively.

Now, consider the group-wise target PageRank score, defined as $\phi_1 = \frac{\gamma^2-3\gamma+3}{3(2-\gamma)}$ and $\phi_2 = 1 - \phi_1$. It holds that $\frac{1}{2}L(\hat{\mathbf{P}}_1, \gamma, \mathbf{v}) + \frac{1}{2}L(\hat{\mathbf{P}}_2, \gamma, \mathbf{v}) = 0$. One can easily find a γ such that that $L(\frac{1}{2}\hat{\mathbf{P}}_1 + \frac{1}{2}\hat{\mathbf{P}}_2, \gamma, \mathbf{v}) = L(\hat{\mathbf{P}}_3, \gamma, \mathbf{v}) > 0$, which contradicts the definition of convexity. This completes the proof.

B Proof of Proposition 2

Proposition 2. *The gradient of the fairness-loss function $L(\mathbf{P}, \gamma, \mathbf{v})$ with respect to the transition matrix $\mathbf{P}$ is given by*

$$\frac{\partial}{\partial \mathbf{P}} L(\mathbf{P}, \gamma, \mathbf{v}) = \frac{2(1-\gamma)}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right) \mathbf{p}\mathbf{y}_k^\top, \tag{4.1}$$

where $\mathbf{p}$ is the PageRank vector with respect to $(\mathbf{P}, \gamma, \mathbf{v})$, and $\mathbf{y}_k = \mathbf{U}^{-T}\mathbf{1}_k = (\mathbf{I} - (1-\gamma)\mathbf{P})^{-1}\mathbf{1}_k$, with $\mathbf{U} = \mathbf{I} - (1-\gamma)\mathbf{P}^\top$.

Recalling that $\mathbf{p} = \gamma\mathbf{U}^{-1}\mathbf{v}$, we have the following:

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{P}} L(\mathbf{P}, \gamma, \mathbf{v}) &= \frac{\partial}{\partial \mathbf{P}} \frac{1}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right)^2 \\
&= \frac{2}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right) \\
&= \frac{2}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{p}\right) \\
&= \frac{2\gamma}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v}\right).
\end{aligned}$$

By the chain rule of matrix calculus [28, Equation (137)], we obtain:

$$\frac{\partial}{\partial \mathbf{P}[i,j]} \left(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v}\right) = \mathrm{Tr}\left[\left(\frac{\partial(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v})}{\partial \mathbf{U}}\right)^\top \frac{\partial \mathbf{U}}{\partial \mathbf{P}[i,j]}\right] \tag{B.1}$$

By considering the derivative of inverse [28, Equation (61)] we have:

$$\frac{\partial(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v})}{\partial \mathbf{U}} = -\mathbf{U}^{-T}\mathbf{1}_k\mathbf{v}^\top\mathbf{U}^{-T}.$$

The gradient of $\mathbf{U}$ with respect to $\mathbf{P}[i,j]$ is:

$$\frac{\partial \mathbf{U}}{\partial \mathbf{P}[i,j]} = -(1-\gamma)\frac{\partial \mathbf{P}^\top}{\partial \mathbf{P}[i,j]} = -(1-\gamma)\mathbf{E}_{ji},$$

where $\mathbf{E}_{ij}$ is the matrix whose $[i,j]$ entry is 1 and all other entries are 0. From Equation (B.1), we get:

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{P}[i,j]} \left(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v}\right) &= (1-\gamma)\mathrm{Tr}\left[\left(\mathbf{U}^{-T}\mathbf{1}_k\mathbf{v}^\top\mathbf{U}^{-T}\right)^\top \mathbf{E}_{ji}\right] \\
&= (1-\gamma)\left(\mathbf{U}^{-1}\mathbf{v}\mathbf{1}_k^\top\mathbf{U}^{-1}\right)[i,j] \\
&= \frac{1-\gamma}{\gamma}\left(\mathbf{p}\mathbf{1}_k^\top\mathbf{U}^{-1}\right)[i,j].
\end{aligned}$$

Substituting (B) into Equation (B) leads to

$$\frac{\partial}{\partial \mathbf{P}} L(\mathbf{P}, \gamma, \mathbf{v}) = \frac{2(1-\gamma)}{K} \sum_{k=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p} - \phi_k\right)\left(\mathbf{p}\mathbf{1}_k^\top\mathbf{U}^{-1}\right) \tag{B.2}$$

C Proof of Proposition 3

Proposition 3. *The gradient of the fairness-loss function $L_g(\mathbf{P}, \gamma)$ with respect to the transition matrix $\mathbf{P}$ is given by*

$$\frac{\partial}{\partial \mathbf{P}} L_g(\mathbf{P}, \gamma) = \frac{2(1-\gamma)}{K^2} \sum_{k=1}^{K}\sum_{\ell=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right)\left(\mathbf{p}_\ell\mathbf{1}_k^\top\mathbf{U}^{-1}\right). \tag{4.2}$$

where $\mathbf{p}_\ell$ is the PageRank vector with respect to $(\mathbf{P}, \gamma, \mathbf{v}_\ell)$.

Similarly to the proof in Appendix B, for the group-adapted fairness loss, we can compute its gradient over $\mathbf{P}$ as follows:

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{P}} L_g(\mathbf{P}, \gamma) &= \frac{\partial}{\partial \mathbf{P}} \frac{1}{K^2} \sum_{k=1}^{K}\sum_{l=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right)^2 \\
&= \frac{2}{K^2} \sum_{k=1}^{K}\sum_{l=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right) \\
&= \frac{2}{K^2} \sum_{k=1}^{K}\sum_{l=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{p}_\ell\right) \\
&= \frac{2\gamma}{K^2} \sum_{k=1}^{K}\sum_{l=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right) \frac{\partial}{\partial \mathbf{P}} \left(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v}_\ell\right).
\end{aligned} \tag{C.1}$$

According to Appendix B,

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{P}[i,j]} \left(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{v}_\ell\right) &= (1-\gamma)\left(\mathbf{U}^{-1}\mathbf{v}_\ell\mathbf{1}_k^\top\mathbf{U}^{-1}\right)[i,j] \\
&= \frac{1-\gamma}{\gamma}\left(\mathbf{p}_\ell\mathbf{1}_k^\top\mathbf{U}^{-1}\right)[i,j].
\end{aligned} \tag{C.2}$$

Substituting Equation (C.2) to Equation (C.1), we get

$$\frac{\partial}{\partial \mathbf{P}} L_g(\mathbf{P}, \gamma) = \frac{2(1-\gamma)}{K^2} \sum_{k=1}^{K}\sum_{\ell=1}^{K} \left(\mathbf{1}_k^\top \mathbf{p}_\ell - \phi_k\right)\left(\mathbf{p}_\ell\mathbf{1}_k^\top\mathbf{U}^{-1}\right) \tag{C.3}$$

D Approximating $\mathbf{y}_k$

In this section, we prove that for approximating $\mathbf{y}_t$, truncating the Neumann series to its first 50 terms yields a relative error below 0.0003.

According to the Neumann series formula, we can calculate the matrix inverse of a matrix $\mathbf{I}-\mathbf{X}$ as: $(\mathbf{I}-\mathbf{X})^{-1} = \sum_{t=0}^{\infty}\mathbf{X}^{\mathsf{T}}$. Let $\mathbf{S}_t$ denote the partial sum of the Neumann series up to the t-th term, that is,

$$\mathbf{S}_t = \sum_{i=0}^{t}\mathbf{X}^{\mathsf{T}} = \mathbf{I}+\mathbf{X}+\mathbf{X}^2+\cdots+\mathbf{X}^k.$$

We denote the truncation error as $\mathbf{E}_t = (\mathbf{I}-\mathbf{X})^{-1}-\mathbf{S}_t$, then we can express $\mathbf{E}_t$ as the sum of the remaining terms:

$$\mathbf{E}_t = \sum_{i=t+1}^{\infty}\mathbf{X}^{\mathsf{T}}$$

By factoring out $\mathbf{X}^{t+1}$ from the series we get:

$$\mathbf{E}_t = \mathbf{X}^{t+1}\sum_{i=0}^{\infty}\mathbf{X}^{\mathsf{T}}.$$

Using the formula for the Neumann series, the infinite sum $\sum_{t=0}^{\infty}\mathbf{X}^{\mathsf{T}}$ is equal to $(\mathbf{I}-\mathbf{X})^{-1}$. Thus:

$$\mathbf{E}_t = \mathbf{X}^{t+1}(\mathbf{I}-\mathbf{X})^{-1}$$

By taking the matrix norm of both sides, we get

$$\|\mathbf{E}_t\| \le \|\mathbf{X}^{t+1}\|\cdot\|(\mathbf{I}-\mathbf{X})^{-1}\|$$

Given that $|\mathbf{X}|<1$, it follows that

$$\|\mathbf{E}_t\| \le \|\mathbf{X}\|^{t+1}\cdot\|(\mathbf{I}-\mathbf{X})^{-1}\|$$

Thus, we conclude that the relative error for approximating $(1-\mathbf{X})^{-1}$ using the first t term of the Neumann series is given by $\|\mathbf{X}\|^{t+1}$.

Since $\mathbf{U}^{\mathsf{T}} = 1-(1-\gamma)\mathbf{P}$, to evaluate the approximation error of $\mathbf{y}_t = \mathbf{U}^{-T}\mathbf{1}_k$, we set $\mathbf{X} = (1-\gamma)\mathbf{P}$ with $\gamma = 0.15$. Consequently, $\|(1-\gamma)\mathbf{P}\|_\infty = 0.85$. It follows that the relative error of the t-term approximation is 0.85^{t+1}. Setting $t = 50$ yields a relative error below 0.0003.

E Omitted algorithms

In this section, we provide pseudocode for Algorithms 2 and 3.

Algorithm 2 invokes Algorithm 4 as a subroutine to compute the optimal Lagrangian dual variable λ^*. The implementation in Algorithm 4 differs slightly in form from the procedure described by Adam and Mácha [1], yet both are mathematically equivalent. For completeness, we now present a formal justification of the correctness of Algorithm 4, establishing that it indeed returns the optimal dual variable.

According to Adam and Mácha [1], the solution to the projection defined in Section 4.1 is $s_i^* = \mathrm{clip}(s_i+\lambda^*, \ell_i, u_i)$, with $\mathrm{clip}(x,a,b)$ defined as

$$\mathrm{clip}(x,a,b) = \begin{cases} a & \text{if } x \le a, \\ x & \text{if } a < x < b, \\ b & \text{if } x \ge b, \end{cases}$$

and λ^* is the solution to the following equation

$$h(\lambda) := \sum_{i=1}^{K}\mathrm{clip}(s_i+\lambda, \ell_i, u_i) - 1 = 0.$$

Since h is piecewise linear and non-decreasing in λ, we can find λ^* via a simple method proposed by Adam and Mácha [1]:

(1) For any λ, $\mathrm{clip}(s_i, \ell_i, u_i)$ takes value ℓ_i, $\lambda+s_i$, or u_i, thus, we can rewrite $h(\lambda)$ as $h(\lambda) = \mathrm{Active}(\lambda)\cdot\lambda + C$ for some constant C, where $\mathrm{Active}(\lambda)$ is the number of elements such that $s_i+\lambda$ falls in range $[\ell_i, u_i)$.
(2) Let R be the increasing ranking of $2N$ items consisting of $\boldsymbol{\ell}-\mathbf{s}$ and $\mathbf{u}-\mathbf{s}$. For any λ in interval $[R_j, R_{j+1})$, the function $\mathrm{Active}(\lambda)$ remains constant.
(3) Let $\mathrm{Total}(\lambda) = \sum_{i=1}^{N}\mathrm{clip}(s_i+\lambda, \ell_i, u_i)$ and assume that $\mathcal{F}_2$ is non-empty. Then there must exist some $j \in \{1,\cdots,2N\}$ such that $\mathrm{Total}(R_j) \le 1$ and $\mathrm{Total}(R_{j+1}) > 1$, and we can conclude that λ^* lies in range $[R_j, R_{j+1})$. Since $\mathrm{Active}(R_j) = \mathrm{Active}(\lambda^*)$, it follows that $(\lambda^*-R_j)\cdot\mathrm{Active}(R_j) = 1-\mathrm{Total}(R_j)$ and thus we can compute λ^* as

$$\lambda^* = \frac{1-\mathrm{Total}(R_j)}{\mathrm{Active}(R_j)}.$$

Algorithm 3 `AdaptGD`: Projected gradient descent for Group-adapted $\boldsymbol{\phi}$-PageRank and Restricted Group-adapted $\boldsymbol{\phi}$-PageRank

Input: $\mathbf{P}$, γ, $\mathbf{v}$, $\boldsymbol{\phi} = (\phi_1,\ldots,\phi_K)$, group indicator vectors $\mathbf{1}_k$ for $k = \{1,\cdots,K\}$, learning rate α, iteration exponent t_1 and t_2, convergence criterion κ, maximum number of iterations N_{iter}, modification bound δ and ϵ (optional).
Output: Optimized transition matrix $\hat{\mathbf{P}}$

```
P̂ ← P, iter ← 0, L_prev ← ∞, p = (1/n)1
while |L_g(P̂, γ) − L_prev| > κ and iter < N_iter do
    iter ← iter + 1
    L_prev ← L_g(P̂, γ, v)
    for ℓ = 1 to K do
        p ← iterating p^T = (1 − γ)p^T P̂ + (γ/|V_ℓ|) 1_ℓ^T for t_1 times
        for k = 1 to K do
            y_k ← Σ_{i=1}^{t_2} (1 − γ)^i P̂^i 1_k
            P̂ ← P̂ − α (2(1−γ)/K^2) (1_k^T p_ℓ − φ_k) p_ℓ y_k
        end for
    end for
    P̂ ← Project(P̂, δ, ε)
end while
return P̂
```

F Convergence Analysis for minimizing $L(\mathbf{P})$

In this section, we first obtain an upper bound for the Lipschitz constant for the loss function, and then prove that the projected gradient descent method converges to a stationary point.

Given restart probability γ and restart vector $\mathbf{v}$, we can rewrite the loss function as :

$$L(\mathbf{P}) = \frac{1}{K}\sum_{k=1}^{K}\left(g_k(\mathbf{P})\right)^2$$

where $g_k(\mathbf{P}) = \mathbf{1}_k^{\mathsf{T}}\mathbf{p}-\phi_k$ and $\mathbf{p} = \gamma(\mathbf{I}-(1-\gamma)\mathbf{P}^{\mathsf{T}})^{-1}\mathbf{v}$.

Algorithm 4 `Root-Finding`: find the optimal dual variable

Input: vector $\mathbf{s}$, lower bound vector $\boldsymbol{\ell}$, upper bound vector $\mathbf{u}$
Output: λ^*

```
K = |s|
λ ← non-decreasing ranking of ℓ − s and u − s.
Active = 1, Total ← Σ_j ℓ_j
for i = 2, ⋯, 2K do
    Total ← Total + Active × (λ_i − λ_{i−1})
    if Total ≥ 1 then
        return λ* = (1−Total)/Active + λ_i
    else
        if λ_i belongs to ℓ − s then Active++ else Active−−
    end if
end for
```

F.1 Gradient of the Loss $L(\mathbf{P})$

Though we have already provided the first-gradient analysis of the loss function $L(\mathbf{P})$, we rewrite it here for ease of analysis. We first compute the gradient of L with respect to $\mathbf{P}_{ij}$:

$$\frac{\partial L}{\partial \mathbf{P}_{ij}} = \frac{2}{K}\sum_{k=1}^{K} g_k \frac{\partial h_k}{\partial \mathbf{P}_{ij}}$$

where $h_k(\mathbf{P}) = \mathbf{1}_k^\top \mathbf{p}$.

According to Equation (B.2), $\frac{\partial h_k}{\partial \mathbf{P}} = (1-\gamma)\mathbf{p}\mathbf{1}_k^\top \mathbf{U}^{-1}$, therefore,

$$\frac{\partial h_k}{\partial \mathbf{P}_{ij}} = (1-\gamma)\mathbf{p}_i(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j)$$

where $\mathbf{e}_j$ is the standard basis vector (1 at position j, zero elsewhere).

F.2 Second Derivative of the Loss $L(P)$

We now compute the second derivative of the loss:

$$\begin{aligned}\frac{\partial^2 L}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}} &= \frac{\partial}{\partial \mathbf{P}_{ab}}\left(\frac{\partial L}{\partial \mathbf{P}_{ij}}\right)\\ &= \frac{\partial}{\partial \mathbf{P}_{ab}}\left[\frac{2}{K}\sum_{k=1}^{K} g_k \frac{\partial h_k}{\partial \mathbf{P}_{ij}}\right]\\ &= \frac{2}{K}\sum_{k=1}^{K}\left[\frac{\partial g_k}{\partial \mathbf{P}_{ab}}\frac{\partial h_k}{\partial \mathbf{P}_{ij}} + g_k \frac{\partial^2 h_k}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}}\right]\end{aligned}$$

We calculate each term of the second derivative respectively.

First term:

$$\frac{\partial g_k}{\partial \mathbf{P}_{ab}} = \frac{\partial h_k}{\partial \mathbf{P}_{ab}} = (1-\gamma)\mathbf{p}_a(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_b)$$

Second term:

$$\begin{aligned}\frac{\partial^2 h_k}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}} &= (1-\gamma)\frac{\partial}{\partial \mathbf{P}_{ab}}\left[\mathbf{p}_i(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j)\right]\\ &= (1-\gamma)\left[\frac{\partial \mathbf{p}_i}{\partial \mathbf{P}_{ab}}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) + \mathbf{p}_i\frac{\partial}{\partial \mathbf{P}_{ab}}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j)\right]\end{aligned}$$

We have:

$$\frac{\partial \mathbf{p}_i}{\partial \mathbf{P}_{ab}} = (1-\gamma)[\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}]_i$$

and

$$\frac{\partial}{\partial \mathbf{P}_{ab}}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) = (1-\gamma)\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{U}^{-1}\mathbf{e}_j$$

Combining the two terms, we get

$$\frac{\partial^2 h_k}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}} = (1-\gamma)^2\Big([\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}]_i \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) + \mathbf{p}_i \cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{U}^{-1}\mathbf{e}_j\Big)$$

Thus, the Hessian entry of $L(\mathbf{P})$ at coordinate $((ij),(ab))$ is:

$$\begin{aligned}\frac{\partial^2 L}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}} = \frac{2}{K}\sum_{k=1}^{K}\Bigg[&(1-\gamma)^2\mathbf{p}_a\mathbf{p}_i(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_b)(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) \qquad \text{(F.1)}\\ &+ g_k(\mathbf{P})\cdot(1-\gamma)^2\Big([\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}]_i \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) + \mathbf{p}_i \cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{U}^{-1}\mathbf{e}_j\Big)\Bigg]\end{aligned}$$

F.3 Lipschitz Constant for $\frac{\partial L}{\partial \mathbf{P}}$

Since $L(\mathbf{P})$ is twice differentiable, the first derivative of $L(\mathbf{P})$ is Lipschitz smooth with constant C if and only if the operator norm of the Hessian matrix $\frac{\partial^2 L}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}}$ is bounded by C for all $\mathbf{P}$ in the domain. In this section, we flat the Hessian matrix and show an upper bound for its operator norm, which serves as an upper bound to its Lipschitz constant.

In order to flatten the Hessian matrix, let $r \in [n^2]$ be the row indexes and $s \in [n^2]$ be the column indexes. Given any (i,j) pair, we can calculate a unique flattened row index $r = (i-1)n + j$ and given any (a,b) pair, we can calculate a unique flattened column index $s = (a-1)n + b$. Conversely, given any pair (r,s) we can calculate the unique corresponding four indices $i(r), j(r), a(s)$ *and* $b(s)$ as follows:

$$i(r) = \lfloor \frac{r-1}{n} \rfloor + 1, j(r) = r - (i(r)-1)n$$

$$a(s) = \lfloor \frac{s-1}{n} \rfloor + 1, b(s) = s - (a(s)-1)n$$

We can then rewrite Equation (F.1) as

$$\begin{aligned}\mathbf{H}_{rs} = \frac{2}{K}\sum_{k=1}^{K}\Bigg[&(1-\gamma)^2\mathbf{p}_{a(s)}\mathbf{p}_{i(r)}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{b(s)})(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{j(r)})\\ &+ g_k(\mathbf{P})\cdot(1-\gamma)^2\Big([\mathbf{U}^{-1}\mathbf{E}_{b(s)a(s)}\mathbf{p}]_{i(r)} \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{j(r)}\\ &+ g_k(\mathbf{P})\cdot(1-\gamma)^2\mathbf{p}_{i(r)} \cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{b(s)a(s)}\mathbf{U}^{-1}\mathbf{e}_{j(r)}\Big)\Bigg]\end{aligned}$$

Let

$$\mathbf{H}_{rs}^{(1)} = \frac{2}{K}(1-\gamma)^2\sum_{k=1}^{K}\mathbf{p}_{a(s)}\mathbf{p}_{i(r)}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{b(s)})(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{j(r)})$$

$$\mathbf{H}_{rs}^{(2)} = \frac{2}{K}(1-\gamma)^2\sum_{k=1}^{K} g_k(\mathbf{P})[\mathbf{U}^{-1}\mathbf{E}_{b(s)a(s)}\mathbf{p}]_{i(r)} \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_{j(r)})$$

$$\mathbf{H}_{rs}^{(3)} = \frac{2}{K}(1-\gamma)^2\sum_{k=1}^{K} g_k(\mathbf{P})\mathbf{p}_{i(r)} \cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{b(s)a(s)}\mathbf{U}^{-1}\mathbf{e}_{j(r)}$$

Then it holds that

$$\mathbf{H}_{rs} = \mathbf{H}_{rs}^1 + \mathbf{H}_{rs}^2 + \mathbf{H}_{rs}^3$$

To be able to bound the operator norm of $\mathbf{H}$, we bound the operator norm of $\mathbf{H}^{(1)}$, $\mathbf{H}^{(2)}$, and $\mathbf{H}^{(3)}$ separately.

Before bounding the operator norm of each term, we first proof $\|\mathbf{U}^{-1}\|_2 \leq 1/\gamma$, which will be used multiple times in the following deriviation. The proof is the result of three observations: (1) the largest eigenvalue of $\mathbf{P}^\top$ is 1, (2) the smallest eigenvalue of $\mathbf{U}$ is γ and (3) the operator norm, $\|\mathbf{U}^{-1}\|_2$, of matrix $\mathbf{U}^{-1}$ is the reciprocal of the smallest absolute eigenvalue of $\mathbf{U}$. We provide details for each observation in the following:

(1) Let $\mathbf{1}$ be an all-one vector, i.e. $\mathbf{1} = (1, 1, ..., 1)^\top$. We can see $\mathbf{1}$ is a eigenvector of $\mathbf{P}$ since $\mathbf{P}\mathbf{1} = 1$. We also know the largest eigenvalue is bounded by the infinity norm, which is $\|\mathbf{P}\|_\infty = \max_i \sum_j |\mathbf{P}_{ij}| = 1$. Thus, the maximum eigenvalue of $\mathbf{P}^\top$, which is equal to the maximum eigenvalue of $\mathbf{P}$, is equal to 1.
(2) Let σ denote each eigenvalue of $\mathbf{P}^\top$, and μ each eigenvalue of $\mathbf{U}$. Since $\mathbf{U} = \mathbf{I} - (1-\gamma)\mathbf{P}^\top$, it holds that $\mu = 1-(1-\gamma)\sigma$. Thus the smallest eigenvalue of $\mathbf{U}$ occurs when $\sigma = 1$, leading to $\mu_{min} = 1 - (1-\gamma) = \gamma$
(3) Combining (1) and (2), we get $\|\mathbf{U}^{-1}\| = \frac{1}{\mu_{min}} = \frac{1}{\gamma}$.

F.3.1 Bounding $\|\mathbf{H}^{(1)}\|_2$. Let $\mathbf{w}_k$ of a vector of length n^2 that depends on group k, specifically,

$$\mathbf{w}_k[r] = (1-\gamma)\mathbf{p}_{i(r)}(\mathbf{1}_k^\top \mathbf{U}^{-1} e_{j(r)})$$

Then we can represent $\mathbf{H}^{(1)}$ as a rank 1 matrix as follow:

$$\mathbf{H}^{(1)} = \frac{2}{K}\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^\top$$

By the subadditivity of the spectral norm, and the identity $\|\mathbf{w}_k\mathbf{w}_k^\top\|_2 = \|\mathbf{w}_k\|_2^2$. We first bound the second norm of $\mathbf{w}_k$, it follows that

$$\begin{aligned}
\|\mathbf{w}_k\|_2^2 &= (1-\gamma)^2 \sum_{i=1}^{n} \mathbf{p}_i^2 \sum_{j=1}^{n} (\mathbf{1}_k^\top \mathbf{U}^{-1} e_j)^2 \\
&= (1-\gamma)^2 \sum_{i=1}^{n} \mathbf{p}_i^2 \|\mathbf{1}_k^\top \mathbf{U}^{-1}\|_2^2 \\
&\overset{(a)}{\leq} \frac{(1-\gamma)^2}{\gamma^2}\|\mathbf{1}_k\|_2^2
\end{aligned}$$

where (a) holds because $\sum_i \mathbf{p}_i^2 \leq 1$, and $\|\mathbf{U}^{-1}\| \leq \frac{1}{\gamma}$. We can then bound $\|\mathbf{H}^{(1)}\|_2$ as follow:

$$\|\mathbf{H}^{(1)}\|_2 = \frac{2}{K}\sum_{k=1}^{K}\|\mathbf{w}_k\|_2^2 \tag{F.2}$$

$$\leq \frac{2(1-\gamma)^2}{K\gamma^2}\sum_{k=1}^{K}\|\mathbf{1}_k\|_2^2 = \frac{2(1-\gamma)^2}{K\gamma^2}n \tag{F.3}$$

F.3.2 Bounding $\|\mathbf{H}^{(2)}\|_2$. Let $\mathbf{X} \in \mathbb{R}^{n\times n}$ be a matrix, and let $\mathbf{x} \in \mathbb{R}^{n^2}$ denote the vector obtained by flattening $\mathbf{X}$. We can then obtain a vector $\mathbf{y} = \mathbf{H}^{(2)}\mathbf{x} \in \mathbb{R}^{n^2}$, which, after reshaping, is equal to a matrix $\mathbf{Y} \in \mathbb{R}^{n\times n}$. Specifically,

$$\begin{aligned}
\mathbf{Y}_{i(r),j(r)} = \mathbf{y}_r &= \sum_{s=1}^{n^2} \mathbf{H}^{(2)}_{r,s}\mathbf{x} \\
&= \sum_{a(s)=1}^{n}\sum_{b(s)=1}^{n} \mathbf{H}^{(2)}_{(i(r),j(r)),(a(s),b(s))}\mathbf{X}_{a(s),b(s)} \\
&\overset{(a)}{=} \frac{2(1-\gamma)^2}{K}\sum_{k=1}^{K} g_k \left[\sum_{a,b}[\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}]_i \mathbf{X}_{ab}\right](\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) \\
&= \frac{2(1-\gamma)^2}{K}\sum_{k=1}^{K} g_k \left[\sum_b (\mathbf{U}^{-1})_{i,b}\left(\sum_a \mathbf{p}_a \mathbf{X}_{a,b}\right)\right](\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j)
\end{aligned}$$

where for equality (a) and the following deriviation, we replace $a(s), b(s)$ with a, b, and $i(r), j(r)$ with (i, j) for ease of notation whenever the context is clear.

Let $\mathbf{u} = \mathbf{X}^\top\mathbf{p}$, $\mathbf{z} = \sum_{k=1}^{K} g_k \mathbf{1}_k$, and $\mathbf{w} = \mathbf{A}^{-T}\mathbf{z}$, then we can rewrite $\mathbf{Y}$ as

$$\mathbf{Y} = \frac{2(1-\gamma)^2}{K}(\mathbf{U}^{-1}u)\mathbf{w}^\top$$

Since $(\mathbf{U}^{-1}u)\mathbf{w}^\top$ is a rank-1 matrix, the spectrual norm and the Frobenius norm of $(\mathbf{U}^{-1}u)\mathbf{w}^\top$ are equal. Thus we can bound the operator norm of $\mathbf{Y}$ as

$$\begin{aligned}
\|\mathbf{Y}\|_F &= \frac{2(1-\gamma)^2}{K}\|\mathbf{U}^{-1}\mathbf{u}\|_2\|\mathbf{w}\|_2 \\
&\leq \frac{2(1-\gamma)^2}{K}(\|\mathbf{U}^{-1}\|_2\|\mathbf{u}\|_2)(\|\mathbf{U}^{-1}\|_2\|\mathbf{z}\|_2) \\
&\overset{(a)}{\leq} \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{u}\|_2\|\mathbf{z}\|_2 \\
&\leq \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{X}^\top\mathbf{p}\|_2 \sum_{k=1}^{K}\|\mathbf{1}_k\|_2|g_k| \\
&\overset{(b)}{\leq} \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{X}\|_F\|\mathbf{p}\|_2 \sum_{k=1}^{K}\|\mathbf{1}_k\|_2 \\
&\overset{(c)}{\leq} \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{X}\|_F\sqrt{n}
\end{aligned}$$

where inequality (a) holds because $\|\mathbf{U}^{-1}\|_2 \leq \frac{1}{\gamma}$, (b) holds because $|g_k| \leq 1$ and (c) holds because $\|\mathbf{p}\| \leq 1$.

Therefore, it holds that

$$\begin{aligned}
\|\mathbf{H}^{(2)}\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = \|\mathbf{Y}\|_F &\leq \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{X}\|_F\sqrt{n} \\
&= \frac{2(1-\gamma)^2}{K\gamma^2}\|\mathbf{x}\|_2\sqrt{n}
\end{aligned}$$

by dividing both side by $\|\mathbf{x}\|_2$ and taking the supremum over $\mathbf{x} \neq \mathbf{0}$, we get

$$\|\mathbf{H}^{(2)}\|_2 \leq \frac{2(1-\gamma)^2}{K\gamma^2}\sqrt{n} \tag{F.4}$$

F.3.3 Bounding $\|\mathbf{H}^{(3)}\|_2$. We can reorganize terms in $\mathbf{H}^{(3)}$ and it holds that

$$\mathbf{H}^{(3)}_{rs} = \frac{2(1-\gamma)^2}{K}\sum_{k=1}^{K} g_k \mathbf{p}_{i(r)}(\mathbf{U}^{-1})_{a(s),j(r)}(\mathbf{z}_k)_{b(s)}$$

where $\mathbf{z}_k = \mathbf{1}_k^\top \mathbf{U}^{-1}$.

Let $(\mathbf{C}_k)_{(i(r),j(r)),(a(s),b(s))} = \mathbf{p}_{i(r)}(\mathbf{U}^{-1})_{a(s),j(r)}(\mathbf{z}_k)_{b(s)}$, we can further rewrite $\mathbf{H}^2$ as follow:

$$\mathbf{H}^{(3)}_{rs} = \frac{2(1-\gamma)^2}{K}\sum_{k=1}^{K} g_k \mathbf{C}_k \tag{F.5}$$

Furthermore, since the three factors $\mathbf{p}_i(r)$, $(\mathbf{U}^{-1})_{a(s),j(e)}$, $(\mathbf{z}_k)_{b(s)}$ depend on disjoint index pairs, $\mathbf{C}_k$ can be represented as a triple outer product and reshaped as

$$\mathbf{C}_k = \mathbf{p} \otimes \mathbf{U}^{-1} \otimes \mathbf{z}_k$$

and thus we can bound the operator norm of $\mathbf{C}_k$ as

$$\begin{aligned}\|\mathbf{C}_k\|_2 &= \|\mathbf{p}\|_2 \|\mathbf{U}^{-1}\|_2 \|\mathbf{z}_k\|_2 \\ &\le 1 \times \frac{1}{\gamma} \times \|\mathbf{U}^{-1}\|_2 \|1_k\|_2 \\ &\le \frac{1}{\gamma^2}\|1_k\|_2\end{aligned}$$

Therefore, we can bound the operator norm of $\mathbf{H}^{(3)}$ as

$$\begin{aligned}\|\mathbf{H}^{(3)}\|_2 &\le \frac{2K(1-\gamma^2)}{\gamma^2}\sum_{k=1}^{K}\|1_k\|_2 \\ &= \frac{2K(1-\gamma^2)}{\gamma^2}\sum_{k=1}^{K}\sqrt{|V_k|} \\ &\overset{(a)}{\le} \frac{2K(1-\gamma^2)}{\gamma^2}\sqrt{K}\sqrt{\sum_{k=1}^{K}|V_k|} \\ &= \frac{2K(1-\gamma^2)}{\gamma^2}\sqrt{K}\sqrt{n} \\ &= \frac{2(1-\gamma^2)}{\gamma^2}\sqrt{\frac{n}{K}}\end{aligned} \tag{F.6}$$

where inequality (a) is due to Cauchy-Schwarz inequatlity.

F.3.4 The Lipschitz constant for the first derivative of $L(\mathbf{P})$. Combining Equation (F.2), Equation (F.4) and Equation (F.6), it follows that

$$\|\mathbf{H}\|_2 \le \|\mathbf{H}^{(1)}\|_2 + \|\mathbf{H}^{(2)}\|_2 + \|\mathbf{H}^{(3)}\|_2 \le \frac{2(1-\gamma)^2}{K\gamma^2}(n+\sqrt{n}+\sqrt{Kn})$$

Let $C = \frac{2(1-\gamma)^2}{K\gamma^2}(n+\sqrt{n}+\sqrt{Kn})$, it follows that $\frac{\partial L}{\partial \mathbf{P}}$ is C-Lipschitz.

F.4 Convergence analysis of the projected gradient descent method for minimizing $L(\mathbf{P})$

We have proved $L(P)$ is a differentiable function and its first derivative is C-Lipschitz, since its feasible set $\mathcal{C}_R(\mathbf{P}) \cap \mathcal{C}(\mathbf{P})$ is convex and closed, the projected gradient update with constant step size $0 \le \alpha \le 2/C$ converges to a stationary point, which is a local minimum of $L(\mathbf{P})$ over $\mathcal{C}_R(\mathbf{P}) \cap \mathcal{C}(\mathbf{P})$ [5].

G Convergence Analysis for minimizing $L_g(\mathbf{P})$

We can derive the Hessian matrix of $L_g(\mathbf{P})$ and its Lipschitz constant in a manner analogous to that of $L(\mathbf{P})$. Specifically, let $\mathbf{H}_g$ be the Hessian matrix of L_g, then it holds that

$$\begin{aligned}\mathbf{H}_g[(i,j),(a,b)] &= \frac{\partial^2 L_g}{\partial \mathbf{P}_{ij}\partial \mathbf{P}_{ab}} \\ &= \frac{2}{K^2}\sum_{k=1}^{K}\sum_{l=1}^{K}\Big[(1-\gamma)^2 \mathbf{p}_{\ell a}\mathbf{p}_{\ell i}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_b)(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) \\ &+ g_k(\mathbf{P})\cdot(1-\gamma)^2\Big([\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}_\ell]_i \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) + \mathbf{p}_{\ell i}\cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{U}^{-1}\mathbf{e}_j\Big)\Big]\end{aligned} \tag{G.1}$$

Equation (G.1) follows directly from Equation (F.1) by replacing $\mathbf{p}$ with $\mathbf{p}_\ell$ throughout, introducing an additional summation over $l \in 1, \dots, K$, and dividing by the extra term K. Let $\mathbf{H}_g^k$ denote the inner sum of $\mathbf{H}_g$, that is,

$$\begin{aligned}\mathbf{H}_g^k = \frac{2}{K}\sum_{l=1}^{K}\Big[&(1-\gamma)^2 \mathbf{p}_{\ell a}\mathbf{p}_{\ell i}(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_b)(\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) \\ &+ g_k(\mathbf{P})\cdot(1-\gamma)^2\Big([\mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{p}_\ell]_i \cdot (\mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{e}_j) + \mathbf{p}_{\ell i}\cdot \mathbf{1}_k^\top \mathbf{U}^{-1}\mathbf{E}_{ba}\mathbf{U}^{-1}\mathbf{e}_j\Big)\Big]\end{aligned}$$

It is straightforward to see that the derivation of an upper bound on the operator norm of $\mathbf{H}$ remains valid for $\mathbf{H}_g^k$ when $\mathbf{p}$ is replaced by $\mathbf{p}_\ell$. That is, $\mathbf{H}$ and $\mathbf{H}_g^k$ share the same operator norm upper bound C. Given the subadditivity property of the operator norm, it holds that

$$\|\mathbf{H}_g\|_2 \le \frac{1}{K}\sum_{k=1}^{K}\|\mathbf{H}_g^k\|_2 = \|\mathbf{H}\|_2 = C$$

Therefore, we have established that the operator norm of L_g is bounded by C, which is equivalent to stating that the Lipschitz constant of its gradient is also bounded by C. Consequently, projected gradient descent converges to a local minimum when using a constant step size $0 \le \alpha \le 2/C$.

H Experimental settings

H.1 Datasets

Books[3] This is an undirected network of political books sold on Amazon. Each edge represents frequent co-purchases of two books. The books are categorized into three groups: liberal, neutral, and conservative. Since the network is undirected, each edge is converted into two directed edges.

Blogs [2]. A directed network of political blogs, where each edge represents a hyperlink between two blogs. The vertices are divided into two groups: liberal and conservative blogs.

MIND [40] A dataset built from user impression logs on Microsoft News. Each log captures a user's impressions of current news articles, along with their reading history. News articles are labeled, and we construct a directed graph from these logs, filtering out

[3] http://www-personal.umich.edu/~mejn/netdata/

news articles with a frequency of fewer than 50 occurrences. A directed edge is created from a historically read news article to a current one if both appear in the same user's log. The two vertex groups correspond to lifestyle-related and society-related news[4].

Twitter [31]. **A** directed network of political retweets, where each edge represents a retweet from one user to another. The vertex groups are left-leaning and right-leaning users.

Slashdot [22]. **A** directed network representing friend/foe relationships between users on Slashdot. Since this dataset lacks group labels, we apply the METIS graph partitioning algorithm [17] to divide the vertices into groups.

H.2 Baselines

We present here a description of the baselines we use in our evaluation. In our experiments, $\hat{\mathbf{P}}$ represents the revised transition matrix, and $\hat{\mathbf{v}}$ represents the revised restart vector for both our algorithms and all the baselines.

$\mathsf{LFPR_U}$ [37]. **A** residual-based algorithm that revises both the restart vector and the transition matrix to achieve perfect $\boldsymbol{\phi}$-fairness in PageRank scores. The intuition is that each node distributes a portion of its PageRank to the under-represented groups, and distributes the remaining portion to its neighbors uniformly.

The revised transition matrix is $\hat{\mathbf{P}} = \mathbf{P}_L + \mathbf{R}$, where:

$$\mathbf{P}_L = \begin{cases} \frac{1-\phi_1}{out_2(i)}, & \text{if } (i,j) \in E \text{ and } i \in L_1 \\ \frac{\phi_1}{out_1(i)}, & \text{if } (i,j) \in E \text{ and } i \in L_2 \\ 0 & \text{otherwise} \end{cases} \tag{H.1}$$

and $out_k(i)$ represents the number of out-neighbors of vertex i that belongs to group k, and L_k denotes the set of vertices whose portion of neighbors that belong to group k is smaller than the fairness target ϕ_k. $\mathbf{R}$ is given by

$$\mathbf{R} = \delta_1 \mathbf{x}^\mathsf{T} + \delta_2 \mathbf{y}^\mathsf{T} \tag{H.2}$$

where $\delta_1[i] = \phi_1 - \frac{(1-\phi_1)out_1(i)}{out_2(i)}$, $\delta_2[i] = (1-\phi_1) - \frac{\phi_1 out_2(i)}{out_1(i)}$. $\mathbf{x}$ and $\mathbf{y}$ are two indicator vectors where $\mathbf{x}(i) = 1$ if $i \in V_1$ and $\mathbf{y}(i) = 1$ if $i \in V_2$.

$\mathsf{LFPR_N}$ [37]. **A** neighborhood-based algorithm that achieves perfect $\boldsymbol{\phi}$-fairness in PageRank scores. The revised transition matrix is given by $\hat{\mathbf{P}} = \sum_{k=1}^{2} \phi_k \mathbf{P}_k$, where

$$\mathbf{P}_k = \begin{cases} \frac{1}{out_k(i)}, & \text{if } (i,j) \in E \text{ and } j \in V_k \\ \frac{1}{|V_k|}, & \text{if } out_k(i) = 0 \text{ and } j \in V_k \\ 0 & \text{otherwise} \end{cases} \tag{H.3}$$

FairWalk [29]. This algorithm reweights the edges such that a random walker has an equal probability of visiting each reachable group and an equal probability of visiting each vertex within each reachable group. FairWalk does not change the restart vector, and the revised transition matrix is given by:

$$\hat{\mathbf{P}}[i,j] = \frac{\phi_{\ell_j} \mathbf{P}[i,j]}{\sum_{m \in V_{\ell_j}} P[i,m] \sum_{k=1}^{K} I_k(i)\phi_k}, \tag{H.4}$$

[4]The original dataset contains 14 labels. We aggregate vertices with the labels {lifestyle, food and drink, sports, health} into group 1, and those with the labels {entertainment, travel, tv, news, finance, movies, autos, music, weather, video} into group 2.

where $I_k(i) = 1$ if vertex i has a out-neighbor that belongs to group k and 0 otherwise.

CrossWalk [20]. This algorithm extends FairWalk by taking into account the distance of each vertex to other groups in the random walk. CrossWalk upweights edges that (1) are closer to the group boundaries or (2) connect different groups in the graph. This algorithm does not change the restart vector, and we refer readers to Algorithm 1 in the paper for the revised transition matrix.

I Additional experimental results

I.1 Ranking Coefficient for the group-wise PageRank weights

In Tables 3 and 4, we report the weighted average Spearman's rank correlation coefficient, $\overline{\rho}$, for all methods on the Blogs and Slashdot datasets, respectively. Results for the Books and Twitter datasets are omitted, as the average coefficients are close to zero across all methods, indicating that none of the methods can preserve the original PageRank rankings within groups.

On the Blogs dataset, `AdaptGD`(0.1, 0.1) achieves the highest correlation coefficient, followed by $\mathsf{LFPR_N}$ and $\mathsf{LFPR_U}$. `FairGD`(0.5, 0.1) and `AdaptGD` obtain the lowest values, while the remaining methods yield comparable performance. On the Slashdot dataset, algorithm `AdaptGD`(0.1, 0.1) again achieves the highest value, followed by `AdaptGD`(0.5, 0.1). Algorithms `AdaptGD`, `FairGD`, and its variants `FairGD`(0.1, 0.1) and `FairGD`(0.5, 0.1) produce comparable results, whereas FairWalk and CrossWalk perform the worst.

I.2 Ranking Coefficient for the edge weights

In Tables 5 to 7, we present the average Spearman's rank correlation coefficient, $\tilde{\rho}$, for each method on the Books, Blogs and MIND dataset. For each node, we compute the Spearman's coefficient between the rankings of its original and updated outgoing edge weights, and then report the average coefficient across all nodes. We omit the results for the datasets Twitter and Slashdot, as the average coefficients are all approximately in the range [0.999–1) across all methods.

FairWalk performs consistently well across all three datasets, achieving the best results on the Books dataset and second-best results on the remaining datasets. `AdaptGD`(0.1, 0.1) and `AdaptGD` (0.5,0.1) attain the second-best performance on the Books and Blogs datasets, and the third-best on the MIND dataset. `FairGD` achieves the optimal outcome on the MIND dataset.

For the two-group datasets, $\mathsf{LFPR_U}$ and $\mathsf{LFPR_N}$ perform noticeably worse than the other methods. This outcome is expected, as both methods substantially modify the underlying graph structure.

In conclusion, FairWalk best preserves the edge weight ranking across datasets, while `AdaptGD`(0.1, 0.1) and `AdaptGD`(0.5, 0.1) also perform very well.

Table 3: Average Spearman's coefficient $\overline{\rho}$ between the PageRank weight of each method for the Blogs dataset

ϕ	FairWalk	CrossWalk	LFPR_U	LFPR_N	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	0.573	0.506	0.558	0.547	0.437	0.423	0.437	0.423	**0.594**	0.561
0.2	0.579	0.527	0.552	**0.591**	0.483	0.488	0.515	0.446	0.587	0.568
0.3	0.568	0.497	0.577	0.593	0.534	0.553	0.542	0.433	**0.598**	0.553
0.4	0.602	0.539	0.566	0.597	0.560	0.555	0.540	0.442	**0.616**	0.551
0.5	0.590	0.513	0.588	0.577	0.562	0.564	0.577	0.451	**0.609**	0.557
0.6	0.576	0.490	0.599	0.579	0.557	0.569	0.570	**0.801**	0.614	0.545
0.7	0.588	0.511	0.604	0.567	0.586	0.574	0.573	**0.754**	0.627	0.570
0.8	0.612	0.502	0.602	**0.613**	0.510	0.489	0.493	0.492	0.604	0.575
0.9	0.584	0.498	0.606	0.617	0.516	0.506	0.501	0.503	**0.620**	0.555

Table 4: Average Spearman's coefficient $\overline{\rho}$ between the PageRank weight of each method for the Slashdot dataset

ϕ	FairWalk	CrossWalk	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	0.123	0.124	0.221	0.215	0.217	0.212	**0.295**	0.244
0.2	0.140	0.131	0.219	0.217	0.216	0.213	**0.312**	0.251
0.3	0.146	0.135	0.245	0.239	0.241	0.238	**0.310**	0.251
0.4	0.135	0.143	0.229	0.231	0.235	0.243	**0.308**	0.271
0.5	0.121	0.120	0.218	0.212	0.213	0.223	**0.296**	0.261
0.6	0.131	0.119	0.221	0.228	0.228	0.211	**0.287**	0.254
0.7	0.129	0.117	0.226	0.229	0.232	0.212	**0.282**	0.248
0.8	0.128	0.117	0.235	0.230	0.235	0.224	**0.280**	0.250
0.9	0.125	0.122	0.230	0.224	0.229	0.224	**0.271**	0.254

Table 5: Average Spearman's coefficient $\tilde{\rho}$ between edge weight rankings for the Books dataset

ϕ	FairWalk	CrossWalk	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	**0.989**	0.974	0.897	0.919	0.889	0.853	0.975	0.976
0.2	**0.989**	0.975	0.976	0.919	0.940	0.855	0.976	0.975
0.3	**0.989**	0.975	0.963	0.946	0.976	0.869	0.977	0.977
0.4	**0.989**	0.974	0.969	0.959	0.964	0.871	0.977	0.977
0.5	**0.989**	0.973	0.965	0.961	0.972	0.862	0.978	0.976
0.6	**0.988**	0.976	0.973	0.945	0.950	0.879	0.979	0.977
0.7	**0.988**	0.975	0.942	0.943	0.932	0.865	0.977	0.977
0.8	**0.988**	0.974	0.921	0.939	0.922	0.865	0.976	0.977
0.9	**0.988**	0.974	0.907	0.938	0.920	0.909	0.977	0.977

Table 6: Average Spearman's coefficient $\tilde{\rho}$ between edge weight rankings on the Blogs dataset

ϕ	FairWalk	CrossWalk	LFPR_U	LFPR_N	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	**0.997**	0.991	-0.249	0.282	0.987	0.987	0.987	0.984	0.996	0.996
0.2	**0.997**	0.991	-0.142	0.284	0.992	0.991	0.992	0.984	0.995	0.996
0.3	**0.997**	0.991	-0.107	0.284	0.995	0.995	0.995	0.986	0.996	0.996
0.4	**0.997**	0.991	-0.074	0.284	0.995	0.995	0.995	0.986	0.996	0.996
0.5	**0.997**	0.991	0.157	0.486	0.995	0.995	0.995	0.988	0.996	0.996
0.6	**0.997**	0.991	0.161	0.486	0.995	0.995	0.995	0.995	0.996	0.996
0.7	**0.997**	0.991	0.186	0.486	0.995	0.995	0.995	0.995	0.996	0.996
0.8	**0.997**	0.991	0.226	0.486	0.991	0.990	0.990	0.984	0.995	0.996
0.9	**0.997**	0.991	0.290	0.486	0.987	0.988	0.987	0.984	0.995	0.996

Table 7: Average Spearman's coefficient $\tilde{\rho}$ between edge weight rankings on the MIND dataset.

ϕ	FairWalk	CrossWalk	LFPR_U	LFPR_N	FairGD	FairGD(0.1, 0.1)	FairGD(0.5, 0.1)	AdaptGD	AdaptGD(0.1, 0.1)	AdaptGD(0.5, 0.1)
0.1	0.999	0.998	0.318	0.842	≈ 1	≈ 1	≈ 1	0.998	0.998	0.999
0.2	0.999	0.998	0.306	0.842	≈ 1	≈ 1	≈ 1	0.998	0.999	0.998
0.3	0.999	0.998	0.306	0.842	≈ 1	≈ 1	≈ 1	0.998	0.999	0.998
0.4	0.999	0.998	0.298	0.842	≈ 1	≈ 1	≈ 1	0.998	0.999	0.999
0.5	0.999	0.998	0.298	0.842	≈ 1	0.999	≈ 1	0.998	0.998	0.998
0.6	0.999	0.998	0.298	0.842	≈ 1	≈ 1	0.999	0.998	0.998	0.998
0.7	0.999	0.998	0.295	0.842	≈ 1	≈ 1	0.999	0.998	0.998	0.998
0.8	0.999	0.998	0.295	0.842	≈ 1	0.999	0.999	0.997	0.998	0.998
0.9	0.999	0.998	0.294	0.842	≈ 1	0.999	0.999	0.996	0.998	0.998